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Generic motion planning for multi-locomotion autonomous vehicles: A spatio-temporal decoupling framework to accommodate multiple maneuvering modes

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HIGHLIGHTS

- A generic framework enables motion planning for multi-locomotion autonomous vehicles.
- Automated decision-making for maneuvering modes and geometric paths.
- Local optimization ensures smooth and stable switching between maneuvering modes.
- The method improves feasibility and flexibility in complex and confined spaces.

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ABSTRACT

Multi-locomotion autonomous vehicles (MLAVs) are becoming promising platforms for unmanned driving due to their multi-mode capabilities. Nevertheless, motion planning for MLAVs remains largely unexplored, as existing paradigms cannot fully exploit their multiple maneuvering modes especially in mode decision and switching. To address this, this study proposes a generic spatio-temporal decoupling framework to provide a systematic solution for motion planning with automatic decision-making and smooth switching of maneuvering modes. The framework hierarchically decomposes the problem into spatial and temporal layers. At the spatial layer, a modified A* search and mixed-integer programming are used to co-optimize the geometric path and maneuvering mode sequence. At the spatio-temporal layer, segmented trajectory generation and local transient optimization for intermediate modes are designed to ensure smooth and stable switching and mitigate planning conservatism. Simulation and experiment results demonstrate that the method significantly improves feasibility, smoothness, and stability over baseline strategies, enabling MLAVs to fully exploit their multi-mode capabilities in complex and narrow scenarios.

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1. Introduction

1.1. Background

In recent years, with the diversification of vehicle driving and steering configurations, multi-locomotion autonomous vehicles (MLAVs) have attracted increasing interest as a promising platform for autonomous driving. As shown in Fig. 1, typical MLAV configurations include four-wheel independent drive and steering (4WID-4WIS), four-wheel independent drive combined with front-wheel steering (4WID-FWS), front-and-rear axle steering with independent drive (4WID-4WS), as well as their multi-axle extensions. The redundant control degrees of freedom provided by multiple actuators enable MLAVs to perform multiple maneuvering modes, such as Ackermann steering mode (ASM), diagonal motion mode (DMM), and zero-radius (in-place) rotation mode (ZRM). Under low-speed conditions, MLAVs can substantially improve their accessibility and maneuverability by appropriately combining different maneuvering modes. This capability enables efficient traversal through constrained, cluttered, and narrow spaces. Under high-speed conditions, these redundant control features can improve stability through active yaw moments, and achieve fault-tolerant motion or limp-home operation via actuator reconfiguration.

Despite these hardware advantages, fully exploiting the motion potential of MLAVs in autonomous driving remains a significant challenge. One of the critical bottlenecks lies in motion planning. Traditionally, motion planning is a fundamental problem focused on generating paths or trajectories [1–3]. Numerous classical motion planning methods have provided well-established tools and theoretical foundations. These approaches can be broadly categorized into three classes: sampling-and-search-based methods, curve-fitting methods, and optimization-based methods [4]. Sampling-and-search-based planners sample in the state or control space and evaluate candidate paths/trajectories via a cost function to select the best solution. Representative methods include A* [5] and its variants [6–8], and rapidly-exploring random trees (RRT) [9–11]. Curve fitting methods typically use specific geometric curves to connect the start and target positions, often guided by intermediate waypoints or reference lines. Representative examples include Reeds-Shepp [12], Bézier [13], and Clothoid [14]. Optimization-based methods formulate planning as an optimal control problem and solve it numerically [15,16]. These methods commonly use solutions from the aforementioned methods as initial guesses to improve computational efficiency.

However, these methods are primarily designed for vehicles with a single, fixed maneuvering mode. Directly applying these single-mode schemes to MLAVs fails to capture the full spectrum of feasible motions offered by multiple maneuvering modes. For MLAVs, motion planning is particularly challenging due to the coexistence of continuous spatio-temporal states and discrete maneuvering modes. Moreover, the complex geometric and kinematic constraints during mode switching make the modeling and optimization of MLAVs much more intractable than their single-mode counterparts. In this work, we focus on motion planning for MLAVs in low-speed and narrow scenarios, aiming to bridge this gap by developing a framework that systematically integrates both mode decision and switching.

1.2. Related work

In light of these complexities, developing an effective motion planning framework for MLAVs hinges on addressing two **additional challenges**: the decision and switching of maneuvering modes.

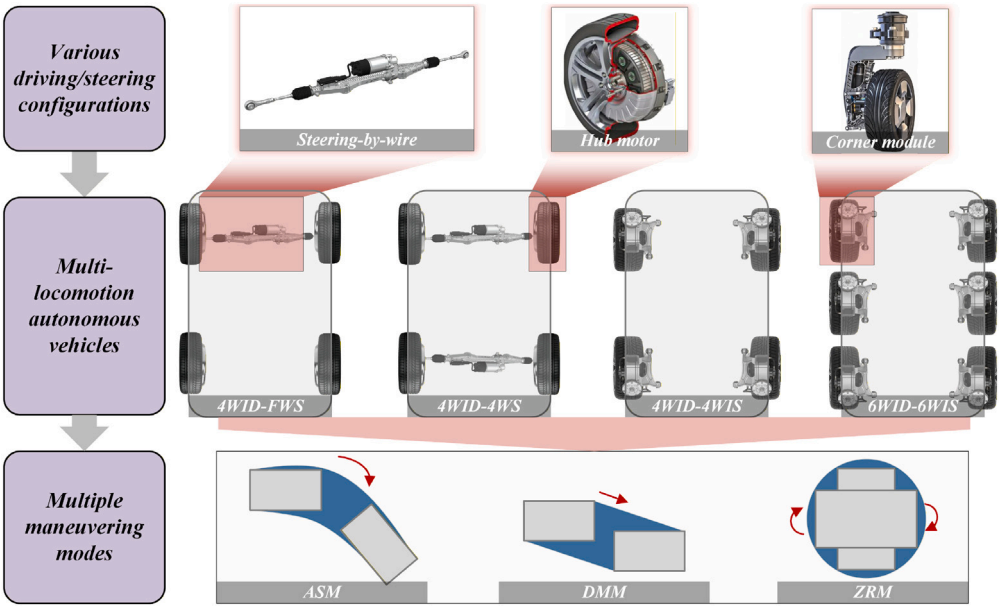


Fig. 1. Representative configurations of multi-locomotion autonomous vehicles.

1.2.1. Decision of maneuvering modes

The main challenge of the decision of maneuvering modes is determining **where** and **when** to switch, and **which** mode to subsequently adopt. It is a complex planning problem that couples continuous spatio-temporal states with discrete event decisions. Consequently, to reduce computational complexity, most prior studies treat trajectory/path planning and the decision of maneuvering modes as separate tasks.

Some studies focus only on trajectory/path planning. For instance, [17] employed Clothoid curves to generate collision-free paths for a 4WID-4WIS vehicle, and subsequently utilized active yaw moment control for path tracking. [18] applied a smoothed A* algorithm to plan a path and designed an iterative speed planning strategy to satisfy both radial and tangential acceleration constraints. [19] developed a Lifelong A*-based method for a differential-wheeled vehicle, accounting for the relationship between the vehicle shape and obstacles during differential steering. While these approaches achieve basic obstacle avoidance, they do not fully exploit multiple maneuvering modes.

Other studies focus exclusively on the decision of maneuvering modes. These approaches usually assume that a path or trajectory has already been planned, and then determine suitable modes through rule-based strategies. [20] designed a fuzzy selector for four-wheel-steered vehicles, using speed, acceleration, and front steering angle as inputs to determine the appropriate control mode. Similarly, [21] classified the maneuvering modes of an omnidirectional mobile robot into ASM, variable Ackermann mode (VAM), DMM, and ZRM. Based on this classification, they further proposed a decision algorithm: when encountering curves exceeding a curvature threshold, the robot switches to ZRM; for other modes, supervisory variables and logical conditions are constructed to guide trial-and-error searches for the optimal sub-controller, while ensuring convergence of a cost function associated with smoothness. [22] further developed a fuzzy selector that takes the yaw rate of the reference trajectory and the quality of the steering motor as inputs to determine the appropriate maneuvering mode.

It is worth emphasizing that the key advantage of MLAVs is their ability to combine multiple maneuvering modes in narrow scenarios and plan trajectories under each mode, thereby significantly improving maneuverability. However, existing methods often treat mode decision and trajectory or path planning as separate tasks, making it impossible to achieve true optimality.

1.2.2. Switching of maneuvering modes

Switching of maneuvering modes addresses **how** to execute a smooth transition once a mode has been selected. It requires transient trajectory planning over the switching interval. Existing studies typically adopt conservative switching strategies oriented toward stability and smoothness, mainly focusing on generating steering angle profiles to improve metrics such as yaw rate, sideslip angle and tire wear. However, the evolution of vehicle position and orientation states is rarely considered in the design of switching strategies. Existing methodologies reported in the literature can be categorized into the following three types.

a) Constant yaw-rate approaches. [23] fit steering-angle profiles with quadratic functions under the constraint of a constant yaw rate to realize switching among front-wheel, rear-wheel and four-wheel steering.

b) Constant turning-radius approaches. For a 4WID-4WIS vehicle, [24] derived the mapping relationship of steering angles at the start and end of switching based on a constant steering radius assumption. For the defined virtual driving wheel, they applied a B-spline curve to fit the steering angle profile and achieved Pareto optimality by optimizing control points. [25] extended this approach to allow switching into DMM by fixing the front-wheel steering angle that the vehicle should reach at the end of the switch. This scheme, however, cannot achieve arbitrary steering configurations in DMM.

c) Instantaneous-center-based approaches. [26] extended prior work by incorporating the target instantaneous center of rotation (ICR) as an optimization variable. They employed particle swarm optimization to solve for the variation trajectory of ICR, thereby enhancing both flexibility and optimality. However, when switching to the DMM, the ICR tends toward infinity. As a result, the endpoint of the B-spline curve cannot be determined, making it impractical to achieve DMM switching in real-world applications.

Although existing strategies enhance smoothness during switching to some extent, they were not originally designed for autonomous driving. Directly applying them to motion planning for MLAVs introduces several limitations. First, the switching criteria employed in these approaches restrict the vehicle's motion freedom during switching, making it difficult to generate accurate and controllable switching trajectories between predefined start and end points. Second, existing transient trajectory planning for mode switching does not adequately account for the evolution of vehicle pose. As a result, they may lead to collisions in constrained environments.

1.3. Motivations and contributions

Existing MLAV motion planning often separates mode decision from trajectory planning, leading to conservative switching strategies and limited flexibility. Based on the above-mentioned points, there is a clear gap between the need to fully exploit MLAVs' multiple maneuvering capabilities and current motion planning paradigms. To bridge this gap, this study proposes a generic hierarchical motion planning framework to provide a systematic solution for motion planning with decision of maneuvering modes and smooth switching of maneuvering modes. Unlike our previous work that focused only on 4WIS vehicles [27], this study generalizes the preliminary concept to MLAVs. Building on our earlier research, the proposed framework introduces spatio-temporal planning and local optimization of transient switching maneuvers, enabling cooperative decision-making for maneuvering modes, trajectory generation, and switching planning.

The main contributions of this study are as follows:

- A spatio-temporal decoupling motion planning framework is proposed to release the motion potential of MLAVs in a structured, layer-by-layer manner. This structure separates the complex planning problem, which involves both continuous states and discrete decisions, into spatial and temporal layers. The spatial layer co-optimizes the geometric path and the decision of maneuvering modes without considering transient switching kinematics. The temporal layer then incorporates timing information to refine the trajectory with detailed motion profiles. This hierarchical framework offers a generic approach to motion planning, mode decision, and switching, enabling full utilization of the flexibility in MLAVs.
- A path planning method integrated with the decision of maneuvering modes is introduced. A modified A* search strategy is developed to generate a coarse path with multiple maneuvering modes, facilitating the generation of collision-avoidance constraints and offering an initial guess for further optimization. A mixed-integer programming (MIP) based path refinement method is then proposed to simultaneously optimize the spatial state sequence and the decision of maneuvering modes. Compared to existing approaches, this method better utilizes the MLAVs' multi-mode kinematic characteristics, overcoming limitations of single-mode planning paradigms.
- A local smooth switching method for maneuvering modes is proposed to mitigate planning conservatism. Two alternative modeling methods are introduced to capture the intermediate kinematic characteristics during switching: one relies on the ICR, while the other employs a simplified single-track model. Both approaches can avoid the numerical singularities discussed by [28]. The switching task is formulated as a local trajectory optimization within a region of interest (ROI). The proposed method significantly enlarges the feasible driving region, enhances switching efficiency, maintains yaw stability and control smoothness, and rigorously adheres to all vehicle constraints. Theoretically, trajectories produced by existing switching strategies constitute a subset of the feasible solutions generated by this method.

2. Kinematic modeling and framework overview

2.1. Kinematic modeling of MLAVs

With reference to the CoG of the vehicle, the maneuvering modes of an MLAV can be broadly categorized as follows: 1) both the position and heading angle change, as in ASM; 2) the heading angle remains unchanged while the position changes, as in DMM; and 3) the position remains fixed while the heading angle changes, as in ZRM. The reference frame may be established at the CoG or at the center of a selected axle, without loss of generality. This paper models the kinematics of MLAVs under multiple maneuvering modes using a switched affine system:

$$\dot{\xi} = \sum_{i=1}^{n_{\mathrm{m}}} \dot{\xi}_{M_i}, \text{ with } \dot{\xi}_{M_i} = \begin{cases} f_{c,M_i}(\xi,u), & \text{if } \alpha_i = 1; \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Wherein M_i belongs to the abstract set $\{M_1, M_2, \ldots, M_{n_{\mathrm{m}}}\}$ and symbolically represents a maneuvering mode, where $i \in I_1^{n_{\mathrm{m}}}$ is the mode index and n_{m} is the total number of such modes available to the MLAV. The control input is u, the system state is ξ, and f_{c,M_i} represents the continuous-time kinematic function for mode M_i. Binary variables are used to represent the current maneuvering mode of the MLAV, i.e.,

$$\alpha_i = 1 \leftrightarrow \text{MLAV is in mode } M_i. \tag{2}$$

Theoretically, the driving and steering commands of all wheels should be treated as control inputs in the kinematic model. Motion planning for MLAVs equipped with multiple steering and driving actuators is challenging due to their multi-input nature. To reduce modeling complexity and improve computational tractability, a common approach is to define one physical or equivalent master steering wheel. Under this simplified model, the steering angles and speeds of the other wheels in each maneuvering mode must be uniquely determined by those of the master wheel through a deterministic mapping, which must be bijective. To ensure invertibility and uniqueness of the mapping, certain reasonable simplifying assumptions are introduced. For instance, the ICR may be constrained to a specific location, such as a line passing through the CoG and perpendicular to the longitudinal axis, to ensure well-behaved mapping computation. While such assumptions may partially limit the MLAVs' local motion freedom, they help reduce complexity and improve computational efficiency at the planning layer. It is worth noting that in subsequent spatio-temporal or switching layer designs of this study, intermediate dynamics and details are reconsidered, thereby compensating for the conservatism introduced by the simplified model and recovering part of the neglected mobility features. The mapping relations of multiple actuators in MLAVs can be generally described as follows:

$$\delta_u = \psi_\delta(\delta_j),\ v_u = \psi_v(v_j), j \in I_{\text{wheel_id}}, \tag{3}$$

where the set $I_{\text{wheel_id}}$ contains the indices of all wheels, and ψ_δ and ψ_v denote the mapping functions for the steering angle and wheel longitudinal velocity, respectively. When a variable serves as a control input, it may be marked with a subscript u. For example, δ_u refers to the steering angle of a master wheel used as a control input.

Running example. We first introduce a 4WID-4WIS vehicle as a running example, with maneuvering modes including ASM, DMM, and ZRM (see Fig. 2). To establish a bijective mapping, the ICR is constrained to lie on the line perpendicular to the longitudinal axis and passing through the CoG for ASM. Without loss of generality, it may also lie on the extensions of the front or rear axles alternatively. For DMM, the ICR is set to coincide with the CoG. The steering angle and acceleration of the rear wheel in the single-track model are chosen as the control inputs, i.e., $u = \begin{bmatrix} \delta_{\mathrm{r}} & a_{\mathrm{r}} \end{bmatrix}^{\mathrm{T}}$. The corresponding mapping relations for the wheel angles and speeds

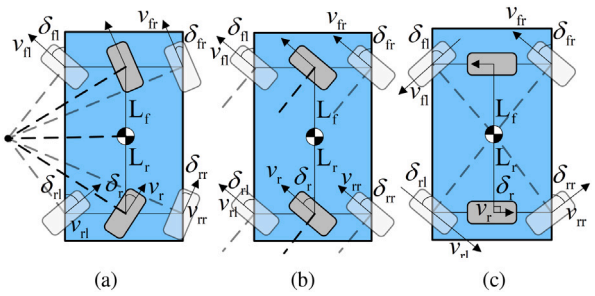


Fig. 2. Kinematic model of a 4WID-4WIS vehicle.

are given as follows:

$$\text{ASM:}\begin{cases}\tan\delta_{\mathrm{fl}}=\dfrac{-2\mathrm{L_f}\tan\delta_{\mathrm{r}}}{2\mathrm{L_r}+\mathrm{B}\tan\delta_{\mathrm{r}}},\ v_{\mathrm{fl}}=-v_{\mathrm{r}}\dfrac{\sin\delta_{\mathrm{r}}}{\sin\delta_{\mathrm{fl}}}\dfrac{\mathrm{L_f}}{\mathrm{L_r}};\\ \tan\delta_{\mathrm{fr}}=\dfrac{-2\mathrm{L_f}\tan\delta_{\mathrm{r}}}{2\mathrm{L_r}-\mathrm{B}\tan\delta_{\mathrm{r}}},\ v_{\mathrm{fr}}=-v_{\mathrm{r}}\dfrac{\sin\delta_{\mathrm{r}}}{\sin\delta_{\mathrm{fr}}}\dfrac{\mathrm{L_f}}{\mathrm{L_r}};\\ \tan\delta_{\mathrm{rl}}=\dfrac{2\mathrm{L_r}\tan\delta_{\mathrm{r}}}{2\mathrm{L_r}+\mathrm{B}\tan\delta_{\mathrm{r}}},\ v_{\mathrm{rl}}=v_{\mathrm{r}}\dfrac{\sin\delta_{\mathrm{r}}}{\sin\delta_{\mathrm{rl}}};\\ \tan\delta_{\mathrm{rr}}=\dfrac{2\mathrm{L_r}\tan\delta_{\mathrm{r}}}{2\mathrm{L_r}-\mathrm{B}\tan\delta_{\mathrm{r}}},\ v_{\mathrm{rr}}=v_{\mathrm{r}}\dfrac{\sin\delta_{\mathrm{r}}}{\sin\delta_{\mathrm{rr}}}.\end{cases}$$

$$\text{DMM: } \delta_{\mathrm{fl}}=\delta_{\mathrm{fr}}=\delta_{\mathrm{rl}}=\delta_{\mathrm{rr}}=\delta_{\mathrm{r}},\ v_{\mathrm{fl}}=v_{\mathrm{fr}}=v_{\mathrm{rl}}=v_{\mathrm{rr}}=v_{\mathrm{r}}. \tag{4}$$

$$\text{ZRM:}\begin{cases}\tan|\delta_{\mathrm{fl}}|=\dfrac{2\mathrm{L_f}}{\mathrm{B}},\ v_{\mathrm{fl}}=-\dfrac{v_{\mathrm{r}}}{\sin\delta_{\mathrm{fl}}}\dfrac{\mathrm{L_f}}{\mathrm{L_r}};\\ \tan|\delta_{\mathrm{fr}}|=\dfrac{2\mathrm{L_f}}{\mathrm{B}},\ v_{\mathrm{fr}}=-\dfrac{v_{\mathrm{r}}}{\sin\delta_{\mathrm{fr}}}\dfrac{\mathrm{L_f}}{\mathrm{L_r}};\\ \tan|\delta_{\mathrm{rl}}|=\dfrac{2\mathrm{L_r}}{\mathrm{B}},\ v_{\mathrm{rl}}=\dfrac{v_{\mathrm{r}}}{\sin\delta_{\mathrm{rl}}};\\ \tan|\delta_{\mathrm{rr}}|=\dfrac{2\mathrm{L_r}}{\mathrm{B}},\ v_{\mathrm{rr}}=\dfrac{v_{\mathrm{r}}}{\sin\delta_{\mathrm{rr}}}.\end{cases}$$

Here, the subscripts for δ and v indicate wheel positions: the first letter denotes front or rear, and the second denotes left or right. The parameters $\mathrm{L_f}$ and $\mathrm{L_r}$ represent the distances from the CoG to the front and rear axles, respectively, L is the wheelbase, and B is the track width. For the 4WID-4WIS vehicle, the state vector is defined as $\xi=\begin{bmatrix}x & y & \varphi & v_{\mathrm{r}}\end{bmatrix}^{\mathrm{T}}$, where (x,y) are the coordinates of CoG, φ is the heading angle, and v_{r} is the virtual rear wheel velocity. The kinematic models under the three maneuvering modes are derived as follows:

$$\dot{\xi}=\begin{cases}f_{c,\mathrm{ASM}}(\xi,u)=\begin{bmatrix}v_{\mathrm{r}}\cos\delta_{\mathrm{r}}\cos\varphi\\ v_{\mathrm{r}}\sin\delta_{\mathrm{r}}\sin\varphi\\ -\dfrac{v_{\mathrm{r}}\sin\delta_{\mathrm{r}}}{\mathrm{L_r}}\\ a_{\mathrm{r}}\end{bmatrix}, & \alpha_1=1;\\ f_{c,\mathrm{DMM}}(\xi,u)=\begin{bmatrix}v_{\mathrm{r}}\cos(\delta_{\mathrm{r}}+\varphi)\\ v_{\mathrm{r}}\sin(\delta_{\mathrm{r}}+\varphi)\\ 0\\ a_{\mathrm{r}}\end{bmatrix}, & \alpha_2=1;\\ f_{c,\mathrm{ZRM}}(\xi,u)=\begin{bmatrix}0 & 0 & -\dfrac{v_{\mathrm{r}}}{\mathrm{L_r}} & a_{\mathrm{r}}\end{bmatrix}^{\mathrm{T}}, & \alpha_3=1.\end{cases} \tag{5}$$

To further illustrate the modeling method for MLAVs, a 4WID-FWS vehicle is also adopted as a running example, with typical maneuvering modes including ASM and ZRM (see Fig. 3). Note that, unlike the 4WID-4WIS configuration, its rotation primarily relies on differential torque between the left and right driving wheels to generate active yaw moment. This rotating motion is closely related to the longitudinal-lateral coupling characteristics of the tires. For example, the yaw rate correlates with the wheel slip ratio and the sideslip angle. To simplify, only kinematic properties with appropriate simplifications are considered at the motion planning layer. Under ZRM, the yaw rate ω can be treated as the primary control variable at the planning level, while the underlying torque distribution and closed-loop controller track this reference ω by generating driving forces or differential torque. The latter falls into the domain of dynamics and control implementation, which is beyond the scope of this study. Considering the above-mentioned points, the main control variable at the planning level may be chosen as $u=\begin{bmatrix}\delta_{\mathrm{f}} & a_{\mathrm{r}} & \omega\end{bmatrix}^{\mathrm{T}}$, and the state is defined as $\xi=\begin{bmatrix}x & y & \varphi & v_{\mathrm{r}}\end{bmatrix}^{\mathrm{T}}$.

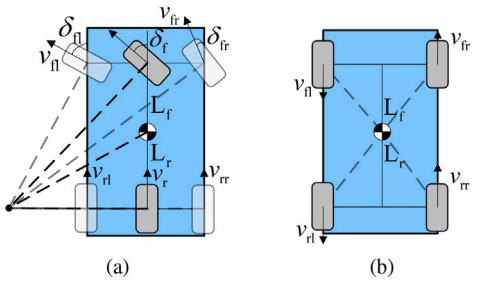


Fig. 3. Kinematic model of a 4WID-FWS vehicle.

Table 1
Motion mode definitions and equivalent control variables for diverse multi-axle configurations.

Chassis configuration	Maneuvering modes	Equivalent control inputs[1]
nWID-nWIS	ASM, DMM, ZRM	δ_r, a_r
nWID-nWS	ASM, DMM, ZRM	δ_r, a_r, ω
nWD-nWS	ASM, DMM	δ_r, a_r
nWID-FWS	ASM, ZRM	δ_f, a_r, ω

[1] Subscripts f and r denote the steering angles of the virtual wheels for the front and rear axles in the equivalent single-track model.

The mapping relations for these two maneuvering modes are as follows:

$$\begin{aligned}
&\text{ASM:}\begin{cases}
\tan\delta_{\mathrm{fl}} = \dfrac{2\mathrm{L}\tan\delta_{\mathrm{f}}}{2\mathrm{L}-\mathrm{B}\tan\delta_{\mathrm{f}}},\ v_{\mathrm{fl}} = \dfrac{v_{\mathrm{r}}\tan\delta_{\mathrm{f}}}{\sin\delta_{\mathrm{fl}}};\\
\tan\delta_{\mathrm{fr}} = \dfrac{2\mathrm{L}\tan\delta_{\mathrm{f}}}{2\mathrm{L}+\mathrm{B}\tan\delta_{\mathrm{f}}},\ v_{\mathrm{fr}} = \dfrac{v_{\mathrm{r}}\tan\delta_{\mathrm{f}}}{\sin\delta_{\mathrm{fr}}};\\
\tan\delta_{\mathrm{rl}} = 0,\ v_{\mathrm{rl}} = \dfrac{v_{\mathrm{r}}(2\mathrm{L}-\mathrm{B}\tan\delta_{\mathrm{f}})}{2\mathrm{L}};\\
\tan\delta_{\mathrm{rr}} = 0,\ v_{\mathrm{rr}} = \dfrac{v_{\mathrm{r}}(2\mathrm{L}+\mathrm{B}\tan\delta_{\mathrm{f}})}{2\mathrm{L}}.
\end{cases}\\
&\text{ZRM:}\delta_{\mathrm{fl}} = \delta_{\mathrm{fr}} = \delta_{\mathrm{rl}} = \delta_{\mathrm{rr}} = 0,\ v_{\mathrm{fl}} = v_{\mathrm{fr}}, v_{\mathrm{rl}} = v_{\mathrm{rr}}.
\end{aligned} \tag{6}$$

The kinematic models for the 4WID-FWS vehicle under the two maneuvering modes are derived as follows:

$$\dot{\xi} = \begin{cases}
f_{c,\mathrm{ASM}}(\xi,u) = \begin{bmatrix} v_{\mathrm{r}}(\cos\varphi - \dfrac{\tan\delta_{\mathrm{f}}\mathrm{L}_r\sin\varphi}{\mathrm{L}}) \\ v_{\mathrm{r}}(\sin\varphi + \dfrac{\tan\delta_{\mathrm{f}}\mathrm{L}_r\cos\varphi}{\mathrm{L}}) \\ \dfrac{v_{\mathrm{r}}\tan\delta_{\mathrm{f}}}{\mathrm{L}} \\ a_{\mathrm{r}} \end{bmatrix}, & \alpha_1 = 1,\\
f_{c,\mathrm{ZRM}}(\xi,u) = \begin{bmatrix} 0 & 0 & \omega & 0 \end{bmatrix}^{\mathrm{T}}, & \alpha_2 = 1.
\end{cases} \tag{7}$$

Remark 1. For multi-axle vehicles with various chassis designs, any two axles can uniquely determine the ICR. Consequently, there is no fundamental difference in kinematics between multi-axle and two-axle vehicles. This principle is further discussed in Section 4.2.2. The proposed modeling framework is fully applicable to multi-axle structures. Table 1 summarizes the motion mode definitions and control input variables for various multi-axle configurations.

2.2. Problem definition and framework overview

Although the aforementioned models for MLAVs are simplified, motion planning based on these models remains challenging. Specifically, how can the optimal maneuvering mode be autonomously selected according to the environment, and how can the efficiency, smoothness, and stability of transient switching between different modes be guaranteed? The objective of this study is to develop a generic motion planning framework for MLAVs that integrates maneuvering mode decision and transient switching planning, thereby fully leveraging the flexibility of MLAVs.

This study adopts a spatio-temporal decoupling framework, as illustrated in Fig. 4. At the spatial planning layer, a coarse path search is first performed to obtain a preliminary guidance path and an initial guess for maneuvering modes, based on which a safe travel corridor is constructed. An MIP-based path optimization model incorporating mode decision is then formulated to derive the optimal maneuvering mode sequence and geometric path within this corridor. At the spatio-temporal planning layer, the mode

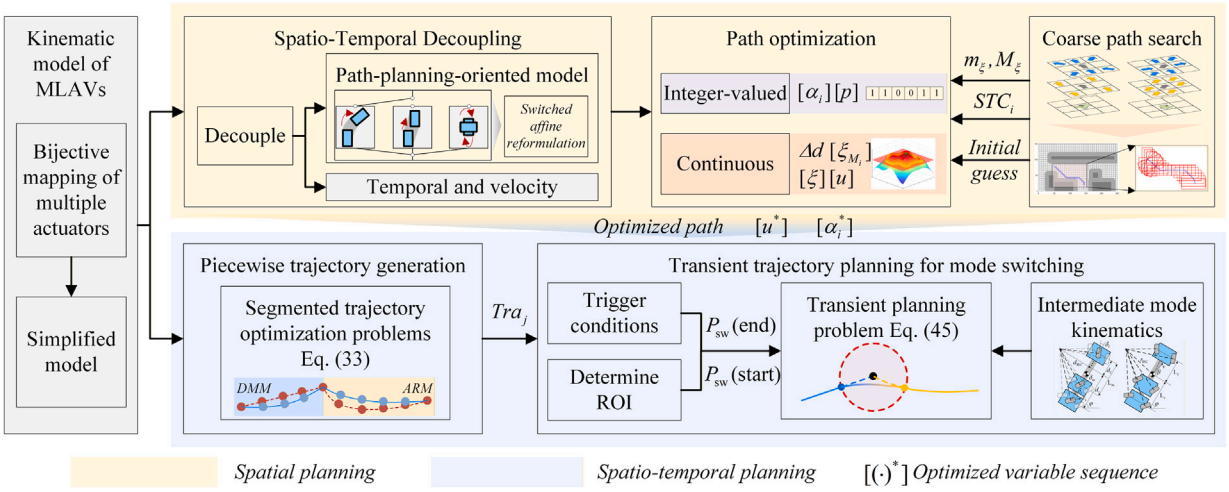


Fig. 4. Overall framework of the proposed motion planning paradigm.

decision is used as a guide to generate segmented time-dependent motion profiles. Finally, transient trajectory planning during mode switching is incorporated to fine-tune the MLAVs' motion within an ROI that optimizes switching efficiency, smoothness, and stability, thereby reducing the conservatism introduced by simplified multi-mode modeling. The detailed implementation steps and the complete workflow are summarized in the pseudo-code provided in Section 4.3.

Remark 2. This study focuses on wheeled MLAVs equipped with trapezoidal steering, hub motors, or corner modules. The proposed framework addresses motion planning in two-dimensional environments. It is noted that tracked-wheeled hybrid platforms are beyond the scope of this work. These platforms typically involve complex mechanical transitions and three-dimensional obstacle avoidance, such as climbing steps or crossing trenches.

3. Spatial planning: coarse search and path optimization

This section presents a spatial planning method that integrates the decisions of multiple maneuvering modes. The approach consists of two main components: coarse path search and path optimization.

3.1. Coarse path search

This study employs a modified A* algorithm for coarse path search to quickly generate a preliminary guide path and an initial guess for maneuvering modes. For MLAVs with multiple maneuvering modes, the main improvements over the standard A* algorithm include:

3.1.1. A multi-mode node expansion strategy

This strategy incorporates the motion characteristics of MLAVs to define expansion rules that account for variations in both position and heading angle. The expansion process is formally described by:

$$\boldsymbol{s}_{\text{child}} = \boldsymbol{s}_{\text{parent}} \oplus (\Delta x, \Delta y, \Delta\varphi)_{M_i}, \tag{8}$$

where $\boldsymbol{s} = (x, y, \varphi)$ represents the node state, $\oplus$ denotes the state composition operator, and $(\Delta x, \Delta y, \Delta\varphi)_{M_i}$ defines the state increments under maneuvering mode M_i, $i \in I_1^{n_{\text{m}}}$.

Running example. The multi-mode node expansion strategies developed for the 4WID-4WIS and 4WID-FWS vehicles are illustrated in Fig. 5. The pose evolution in different maneuvering modes can be coarsely represented in a 3×3 grid.

3.1.2. A node evaluation strategy for different maneuvering modes

This strategy first considers the continuity and feasibility of nodes in adjacent path segments to establish a pruning criterion for inefficient nodes. If a candidate node is found to be inefficient or unreachable, it will be discarded. Otherwise, its quality is evaluated based on a cost function to determine path optimality.

Running example. Take the 4WID-4WIS and 4WID-FWS vehicles as running examples. The ASM may be constrained to prevent continuous changes in heading angle, thereby avoiding excessive curvature. For ZRM, the maneuvering mode sequence is designed to exhibit clustering behavior. This means that consecutive zero-radius rotations are allowed, but the distance between any two adjacent ZRM nodes must exceed a certain threshold to prevent frequent switching. Furthermore, the node evaluation function f is designed to accelerate the search process, trading off some degree of optimality for improved computational efficiency. For instance,

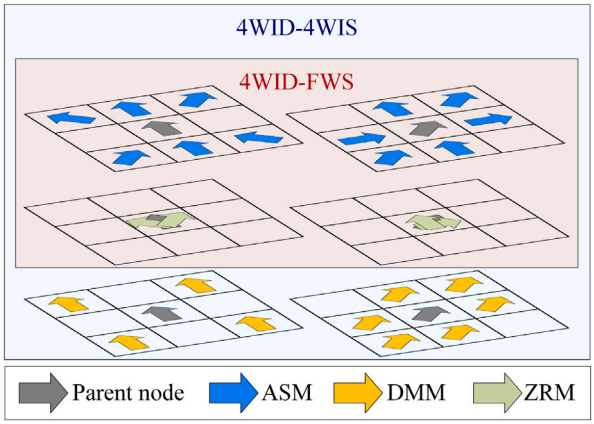


Fig. 5. Multi-mode node expansion strategy for representative configurations.

the accumulated cost g and the heuristic cost h can be defined as follows:

$$\begin{aligned} f &= g + \kappa h, \\ h &= \|s_{\text{child}} - s_{\text{end}}\|_2, \end{aligned} \tag{9}$$

where $\kappa > 1$ encourages faster search convergence, $\|\cdot\|_2$ denotes the Euclidean norm, and s_{end} represents the terminal node. Although this evaluation function may compromise optimality guarantees, the subsequent optimization-based planning layers (as described in Section 3.3) help further improve the path quality.

3.1.3. A relaxed termination condition

To accelerate the search for a feasible path, the termination criterion is relaxed: the search concludes once the vehicle enters a small parking area around the goal, without requiring precise alignment with the target position or heading. As a result, a coarse path $\mathcal{P}_{\text{coarse}}$ and an initial guess of the maneuvering mode sequence $\{M_i\}$ are obtained.

3.2. Redefined modeling by spatio-temporal decoupling

Section 2.1 presented the kinematic models of the vehicle under various maneuvering modes. Directly solving the motion planning problem as a single numerical optimization problem based on this full model poses significant challenges. First, the problem involves both continuous spatio-temporal variables, such as velocity, steering angle, timestamp, and pose, and discrete mode decisions. Second, the transient switching behavior during maneuvering modes is complex. Near switching points, velocity, steering angle, and pose must satisfy specific geometric conditions or human-defined rules. This requirement introduces multiple complicated constraints. To reduce complexity and improve tractability, this study adopts a spatio-temporal decoupling approach. The spatial layer first focuses only on the geometric path and mode decisions, generating a set of time-free path segments and the corresponding mode sequence. The spatio-temporal layer then adds timing and velocity information to these spatial results through segment-wise scheduling, and further optimizes transient switching behaviors locally. This hierarchical decoupling decomposes the originally coupled problem into more tractable sub-problems, without completely separating planning from mode decision as in other existing studies. Based on this spatio-temporal decoupling concept, the following part describes the construction of a redefined path-planning-oriented model.

For general optimization-based motion planning problems, the Euler method is commonly used to discretize the derivatives of the pose states in the kinematic model. Here, ξ generally represents the vehicle state, which includes pose and velocity. Its specific meaning depends on the MLAV configuration and the choice of variables. The total travel time T is often uniformly discretized into $N+1$ sample points, i.e., $\{0, \Delta T, 2\Delta T, \dots, N\Delta T\}$. In this study, we instead adopt a non-uniform discretization strategy by defining time-varying intervals $\Delta T(k)$. The resulting sequence of sampling points is given by $\left\{0, \Delta T'(0), \sum_{i=0}^{1} \Delta T'(i), \dots, \sum_{i=0}^{N-1} \Delta T'(i)\right\}$.

In this paper, δ_u and v_u generally represent the selected steering angle and velocity as control inputs. Since describing changes in the vehicle's pose state is often based on analyzing the velocity components along coordinate axes and rotational motion at a given point, the right-hand side of the above equation necessarily includes the velocity v_u. By extracting the product of velocity and time interval and treating it as a constant value, i.e.,

$$v_u(k) \cdot \Delta T'(k) = p(k) \cdot d/N = p(k) \cdot \Delta d, \tag{10}$$

where d denotes the nominal total travel distance and p indicates the direction of v_u. Both time and velocity can be effectively removed from the kinematic model. Note that this discretization is spatially uniform rather than temporally uniform.

Running example. For the ASM of 4WID-4WIS vehicles, the path-planning-oriented kinematic model is given by

$$\begin{cases} x(k+1) = x(k) + p(k)\cdot \Delta d \cos\delta_{\mathrm{r}}(k)\cos\varphi(k), \\ y(k+1) = y(k) + p(k)\cdot \Delta d \sin\delta_{\mathrm{r}}(k)\sin\varphi(k), \\ \varphi(k+1) = \varphi(k) - p(k)\dfrac{\Delta d \sin\delta_{\mathrm{r}}(k)}{\mathrm{L_r}}. \end{cases} \tag{11}$$

Specifically, for the ZRM of 4WID-FWS vehicles, the simplified model is described as

$$\begin{cases} x(k+1) = x(k), \\ y(k+1) = y(k), \\ \varphi(k+1) = \varphi(k) + p(k)\dfrac{\Delta d}{\sqrt{\mathrm{L_r^2} + \mathrm{B}^2/4}}. \end{cases} \tag{12}$$

Although this model does not explicitly include velocity, it can still be discretized using the spatio-temporally decoupled approach. Assuming v_u represents the actual velocity direction at the center of the rear axle, the angular velocity is derived as $\omega = \dfrac{v_u}{\sqrt{\mathrm{L_r^2} + \mathrm{B}^2/4}}$.

It should be noted, however, that the actual velocity at the rear axle center is closely related to tire slip and sideslip characteristics, meaning the wheel velocities are not mapped in a straightforward manner. The formulas provided here are intended only for kinematic-level spatial path planning and do not address the distribution of wheel speeds in this model. Therefore, the model (12) remains valid under these simplifications.

3.3. Path optimization with decision of maneuvering modes

3.3.1. Kinematic model constraints

The discrete form of the switched affine system (1) is given below:

$$\xi(k+1) = \sum_{i=1}^{n_{\mathrm{m}}} \xi_{M_i}(k+1),\ k \in I_0^{N-1}, \tag{13}$$

with if-then-else conditions:

$$\xi_{M_i}(k+1) = \begin{cases} f_{d,M_i}(\xi(k),u(k)), & \text{if } \alpha_i(k) = 1 \\ 0, & \text{otherwise} \end{cases}, \tag{14}$$

where f_{d,M_i} is the discrete kinematic function for mode M_i. At any time, the vehicle operates in only one maneuvering mode. This is ensured by the following hard constraint:

$$\sum_{i=1}^{n_{\mathrm{m}}} \alpha_i(k) = 1. \tag{15}$$

To formulate the switched affine system as tractable inequality constraints, the big-M method is employed to represent the switching logic:

$$\begin{cases} \xi_{M_i}(k+1) \ \geq m_{\xi}(k+1)\alpha_i(k), \\ \xi_{M_i}(k+1) \ \leq M_{\xi}(k+1)\alpha_i(k), \\ \xi_{M_i}(k+1) - f_{d,M_i}(\xi(k),u(k)) \ \geq -M_{\xi}(k+1)\left(1-\alpha_i(k)\right), \\ \xi_{M_i}(k+1) - f_{d,M_i}(\xi(k),u(k)) \ \leq -m_{\xi}(k+1)\left(1-\alpha_i(k)\right). \end{cases} \tag{16}$$

Here, m_{ξ} and M_{ξ} denote the lower and upper bounds of ξ, respectively. These bounds are derived from the coarse search path:

$$\begin{aligned} &m_x(k) = x_{\mathrm{coarse}}(k) - \beta_x, \quad M_x(k) = x_{\mathrm{coarse}}(k) + \beta_x, \\ &m_y(k) = y_{\mathrm{coarse}}(k) - \beta_y, \quad M_y(k) = y_{\mathrm{coarse}}(k) + \beta_y, \\ &m_{\varphi}(k) = \varphi_{\mathrm{coarse}}(k) - \beta_{\varphi}, \quad M_{\varphi}(k) = \varphi_{\mathrm{coarse}}(k) + \beta_{\varphi}. \end{aligned} \tag{17}$$

Here, $\boldsymbol{\beta} = [\beta_x, \beta_y, \beta_{\varphi}]^{\top}$ represents the confidence level in the preliminary guidance path obtained from the coarse search. Based on the guidance path by the coarse search, a trust region is constructed by expanding it along the three-dimensional pose space (x,y,φ) with a scaling factor controlled by $\boldsymbol{\beta}$. A larger $\boldsymbol{\beta}$ value indicates higher confidence in the preliminary guidance path, resulting in a tighter trust region. This mechanism not only constrains the search and optimization space but also improves convergence and robustness, while reducing computational load and avoiding infeasible solutions.

3.3.2. Actuator saturation constraints

For MLAVs, constraining only the steering angle of the control input does not guarantee that all wheels satisfy the steering angle limits. Taking a 4WID-4WIS vehicle in ASM as an example, when the steering angle of the virtual single-track model is selected as

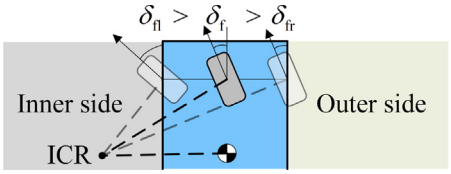


Fig. 6. Comparison between inner and outer wheels under ASM.

the control input, the actual steering angle of the inner wheel is often slightly larger than this virtual value, as shown in Fig. 6. Therefore, to ensure both safety and feasibility, a more reliable and straightforward approach is to impose steering angle constraints that simultaneously apply to all steering wheels. Based on the bijective mapping of steering angles established in Section 2.1, the following constraint is applied:

$$\delta_{\min} \le \left\| \psi_{\delta}^{-1}\left(\delta_u(k)\right) \right\|_{\infty} \le \delta_{\max},\ k \in I_0^N. \tag{18}$$

The symbol ψ_{δ}^{-1} represents the inverse of the mapping function ψ_{δ} defined in (3), which transforms the control input δ_u into a vector of individual wheel steering angles.

3.3.3. Collision avoidance constraints

This study employs the widely used safe travel corridors (STC) method to formulate collision avoidance constraints. Using the preliminary guidance path from the coarse search, a set of corridors is constructed to bound the motion of vehicle feature points. These points are chosen such that disks centered at them collectively cover the outer contour of the vehicle with minimal redundancy. For example, in shorter-wheelbase MLAVs, these may be quartile points along the longitudinal axis. Avoiding collisions is thus approximated by ensuring these disks do not intersect obstacles, which requires all feature points to remain within their respective corridors:

$$C_i^{\text{disc}}(k) \in STC_i(k),\ i \in I_{\text{feature}},\ k \in I_0^N, \tag{19}$$

where I_{feature} denotes the set of all feature point indices, C_i^{disc} denotes a feature point, and STC_i represents the STC generated from C_i^{disc}. These corridors define the feasible region for optimization, allowing the solver to find an optimal path.

3.3.4. Initial and terminal constraints

The following constraints define the MLAV's initial and terminal state conditions:

$$\begin{cases} [x(0), y(0), \varphi(0)] = [x_{\text{start}}, y_{\text{start}}, \varphi_{\text{start}}], \\ \square_{\text{Vehicle_body}}(N) \subset \Gamma_{\text{parking}}, \\ \varphi_{\text{end}} - \gamma < \varphi(N) < \varphi_{\text{end}} + \gamma, \\ \delta_u(N-1) = 0, \end{cases} \tag{20}$$

where $\square_{\text{Vehicle_body}}$ denotes the bounding rectangle of the MLAV's contour, and γ defines the acceptable tolerance between the final vehicle heading and the desired terminal heading φ_{end}. Consistent with the coarse path search method, the terminal pose is not required to be exact. It only needs to lie within a small bounded region Γ_{parking}.

Remark 3. Relaxing strict terminal equality constraints into inequality-based regions is a widely adopted strategy in optimization-based motion planning. As noted in [29], a vehicle fully contained within the designated parking area poses no safety risks. Moreover, a slightly tilted terminal posture can even facilitate the subsequent exit maneuver. From a computational perspective, this approach significantly expands the feasible search space and reduces the numerical burden, particularly for planning problems with finite discretization steps. A detailed analysis of how such relaxations influence MIP solver performance is provided in Section 5.2.

3.3.5. Optimization problem formulation

This study focuses on cost design differences and relevant considerations introduced by multiple maneuvering modes. In the optimization-based trajectory planning community, minimizing travel time is often preferred. Since this study removes time from the path optimization stage, the total path length is naturally considered. As the kinematic modeling and simplifications use the CoG as the reference point, the accumulated displacement of the CoG is chosen to approximate the path length. The corresponding cost function J_{length} is designed as follows:

$$J_{\text{length}} = \sum_{k=0}^{N-1} (x(k+1) - x(k))^2 + (y(k+1) - y(k))^2. \tag{21}$$

The steering rate is a key metric for evaluating control smoothness. It should be noted, however, that the allowable range of the control input δ_u varies significantly across different maneuvering modes. For example, in 4WID-4WIS vehicles, the control input is typically the steering angle of a virtual single-track model. In the ASM, its value does not equal the actual steering angles of the physical wheels. When the inner wheel reaches its limit $\delta_{\max}$, δ_u does not necessarily reach that same limit. In DMM, however, both

the real wheel angles and the virtual input δ_u can simultaneously reach the limit $\delta_{\max}$. This indicates that the steering angle rate should be evaluated in a mode-dependent manner to avoid incorporating meaningless angular increments at mode switching points into the cost or constraints. To address this, mode-specific auxiliary steering angle definitions are introduced for each maneuvering mode, enabling uniform smoothness evaluation and constraint formulation. The detailed definitions are as follows:

$$\delta_{\mathrm{u},M_i}(k) = \begin{cases} \delta_{\mathrm{u}}(k), & \text{if } \alpha_i(k) = 1; \\ 0, & \text{otherwise.} \end{cases} \tag{22}$$

The cost function for the steering rate of the control input is defined as:

$$J_{\text{steer}} = \sum_{i=1}^{n_{\mathrm{m}}} \sum_{k=0}^{N-2} \left(\delta_{\mathrm{u},M_i}(k+1) - \delta_{\mathrm{u},M_i}(k) \right)^2. \tag{23}$$

Running example. Taking the 4WIS-4WID vehicle as a running example, the steering angle range of the control input δ_u in ASM is defined as:

$$-\tan^{-1}\frac{2\mathrm{L_r}\tan\delta_{\max}}{2\mathrm{L_r} + \mathrm{B}\tan\delta_{\max}} \le \delta_{\mathrm{u}} \le \tan^{-1}\frac{2\mathrm{L_r}\tan\delta_{\max}}{2\mathrm{L_r} + \mathrm{B}\tan\delta_{\max}}. \tag{24}$$

In DMM, the range is:

$$-\delta_{\max} \le \delta_{\mathrm{u}} \le \delta_{\max}. \tag{25}$$

For ZRM, the steering input is fixed at 90 degrees. Therefore, both its upper and lower bounds are set to 90, i.e.,

$$\pi/2 \le \delta_{\mathrm{u}} \le \pi/2. \tag{26}$$

Following the approach in (16), these if-then-else conditions (22) can be transformed into tractable inequality constraints using the big-M method.

Frequent switching of maneuvering modes can adversely affect the smoothness of MLAV motion. To reduce unnecessary mode switches, the following cost term is introduced:

$$J_{\text{switch}} = \sum_{i=1}^{n_{\mathrm{m}}} \sum_{k=0}^{N-2} \left(\alpha_i(k+1) - \alpha_i(k) \right)^2. \tag{27}$$

Additionally, a cost term is added to further reduce the number of travel direction reversals:

$$J_{\text{direction}} = \sum_{k=0}^{N-2} \left(p(k+1) - p(k) \right)^2. \tag{28}$$

Cost terms (21), (23), (27), and (28) penalize the path length, steering smoothness, frequency of mode switching, and frequency of travel direction reversals, respectively. The total cost function J_1 can be customized based on vehicle characteristics and environmental requirements, typically formulated as a weighted sum of several performance terms like:

$$J_1 = \eta_{\text{length}} J_{\text{length}} + \eta_{\text{steer}} J_{\text{steer}} + \eta_{\text{switch}} J_{\text{switch}} + \eta_{\text{direction}} J_{\text{direction}}, \tag{29}$$

where the weight coefficients for the corresponding cost terms are specified according to the requirements. For instance, heavy-duty port logistics vehicles typically place greater weight on J_{steer}, J_{switch}, and $J_{\text{direction}}$ to ensure smooth operation. In contrast, small indoor robots may assign higher weight to J_{length} to prioritize fast task response. Based on the above cost and constraints, the path optimization with decision of maneuvering modes is formulated as a MIP, expressed formally as follows:

$$\begin{aligned} &\min_{I,C} \quad J_1 \\ &s.t. \quad \begin{cases} \text{Kinematic constraints of MLAVs;} \\ \text{Actuator saturation constraints;} \\ \text{Collision avoidance constraints;} \\ \text{Initial and Terminal Constraints.} \end{cases} \end{aligned} \tag{30}$$

Here, $\boldsymbol{I}$ represents integer variables, including discrete decisions such as maneuvering modes and travel directions. $\boldsymbol{C}$ denotes continuous variables, including the vehicle's pose states and control inputs. Solving this mixed-integer problem yields a discrete mode sequence $\{M_i^*\}$, an optimized path $\mathcal{P}^*$, and a series of control actions, thereby achieving coordinated optimization of mode decision and path planning within a unified framework.

3.3.6. Tricks for computational efficiency

The primary role of the coarse path search is to generate an initial guide path, which helps to establish collision avoidance constraints and specify the bounds of vehicle pose for the big-M method. Additionally, the preliminary guess of maneuvering modes produced during this process can serve as an initial integer solution for the MIP problem. However, when this initial integer solution is kinematically inconsistent with the vehicle's motion characteristics, its effectiveness in accelerating the solution process is limited.

To further improve solution efficiency, additional cutting planes can be introduced to eliminate inefficient solutions. This helps tighten the feasible region and reduces the gap between the relaxed solution and the optimal solution. Although the cost function already considers the smoothness of the resulting control sequence, hard constraints can still be added to eliminate inefficient solutions. For example, to prevent the vehicle from switching travel directions consecutively, the following constraint can be imposed:

$$\begin{cases} p(k) \geq p(k-1) + p(k+1) - 1, k \in I_1^{N-2}; \\ p(k) \leq p(k-1) + p(k+1), k \in I_1^{N-2}. \end{cases} \tag{31}$$

Furthermore, to encourage the clustering of the same maneuvering mode, the following constraint can be added:

$$\alpha_i(k) \leq \alpha_i(k-1) + \alpha_i(k+1), i \in I_1^{n_\mathrm{m}}, k \in I_1^{N-2}. \tag{32}$$

This constraint ensures that if a certain maneuvering mode is active at the k-th time step, it must also be selected in at least one of the adjacent time steps ($k-1$ or $k+1$).

4. Spatio-temporal planning: trajectory generation and mode switching

4.1. Piecewise trajectory generation

This subsection uses the spatial planning results as a reference to perform segmented spatio-temporal trajectory generation and optimization. This process supplements waypoints with velocity and timing information and refines transient switching behaviors, ensuring kinematic feasibility and smooth transitions.

It is temporarily assumed that the vehicle comes to a stop and resets the steering angle to zero before performing a mode switch. The wheels are then rotated to meet the steering constraints of the target maneuvering mode. This approach is called the stop-and-switch strategy. Denote the total number of mode switches along a continuous path by N_sw. The path can be partitioned into $N_\mathrm{sw}+1$ segments using the path points where mode switches occur. Each segment is assigned an index seg_j, $j \in I_1^{N_\mathrm{sw}+1}$. Every segment contains $N_{\mathrm{seg}_j}+1$ path points, each denoted as $P_{\mathrm{seg}_j}(k)$. Each path point comprises all relevant pose states, control inputs, and other information. A specific quantity within a path point, which may be a scalar or a vector, is accessed via the superscript. For example, the pose information of a path point is written as $P^{\xi}_{\mathrm{seg}_j}(k)$, $k \in I_0^{N_{\mathrm{seg}_j}}$. The segmented trajectory generation strategy fixes the start and end points of each segment and optimizes the trajectory of the MLAV under the corresponding maneuvering mode. Compared to the original path, the number of trajectory points in each segment may be appropriately increased to enrich motion details. This leads to the formulation of $N_\mathrm{sw}+1$ segmented trajectory optimization problems as follows:

$$\min J_2 \tag{33}$$

$$\text{s.t. } \xi(k+1) = f_{d,M_i}\big(\xi(k),u(k)\big), i \in I_1^{n_\mathrm{m}}; \tag{33a}$$

$$\xi(0) = P^{\xi}_{\mathrm{seg}_j}(0), \xi\big(N_{\mathrm{seg}_j}\big) = P^{\xi}_{\mathrm{seg}_j}\big(N_{\mathrm{seg}_j}\big); \tag{33b}$$

$$u(0) = 0, u\big(N_{\mathrm{seg}_j} - 1\big) = 0; \tag{33c}$$

$$C_i^{\mathrm{disc}}(k) \in STC_i(k), i \in I_\mathrm{feature}; \tag{33d}$$

$$\delta_\mathrm{min} \leq \|\psi_\delta^{-1}\big(\delta_u(k)\big)\|_\infty \leq \delta_\mathrm{max}. \tag{33e}$$

Wherein $j \in I_1^{N_\mathrm{sw}+1}$, $k \in I_0^{N_{\mathrm{seg}_j}}$, and the notation $\|\cdot\|_\infty$ denotes the infinity norm. The cost function J_2 can be customized by the user. For example, to optimize total travel time and ensure smooth velocity and steering control, the cost can be defined as

$$J_2 = \eta_\mathrm{time} T + \eta_\mathrm{longitudinal} \sum_{k=0}^{N_{\mathrm{seg}_j}} a^2(k) + \eta'_\mathrm{steer} \sum_{k=1}^{N_{\mathrm{seg}_j}-1} \big(\delta_u(k) - \delta_u(k-1)\big)^2. \tag{34}$$

The weights η_time, $\eta_\mathrm{longitudinal}$, and η'_steer corresponding to these cost terms can be tuned according to requirements, similar to those in (29). For instance, a higher demand for ride comfort would lead to larger values for $\eta_\mathrm{longitudinal}$ and η'_steer.

Constraint (33a) enforces vehicle kinematics, where the model is selected according to the predetermined maneuvering mode. Constraints on the initial and terminal conditions are given by fixing the start and end points of each segment, as specified in (33b) and (33c). Constraint (33d) enforces that all feature points remain within their corresponding safe corridors to avoid collisions. Finally, constraint (33e) imposes steering angle limits for all wheels.

4.2. Transient trajectory planning for mode switching

In Section 2.1, the introduction of ICR-related assumptions helps reduce the complexity involved in planning for multiple actuators. These constraints establish a bijective mapping between the steering angles and speeds of the wheels, allowing the control input to be simplified to a single variable. Although this approach reduces the complexity, it also shrinks the feasible operating region and restricts the MLAV's flexibility. Furthermore, during mode switching, the vehicle must come to a complete stop before repositioning the wheels to satisfy the steering constraints of the target mode. Consequently, the resulting trajectories are often overly conservative

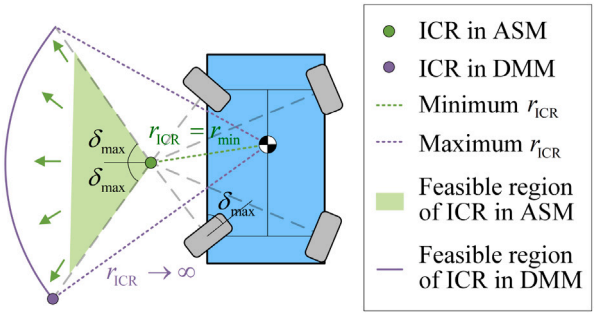


Fig. 7. Feasible region of the ICR for a 4WID-4WIS vehicle.

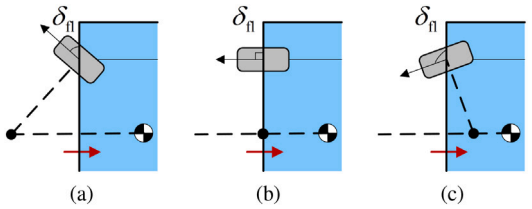


Fig. 8. Schematic of the singularity during a direct X-to-ZRM switching.

and do not fully exploit the vehicle's maneuvering capabilities in multi-mode operation. To overcome these limitations, this subsection analyzes the motion characteristics during mode switching and introduces a dedicated transient planning method.

The feasible region of the ICR is determined by the steering angle limits of the MLAV. Taking the 4WID-4WIS vehicle as an example, the ICR for both ASM and DMM lies outside the contour of the vehicle, whereas for ZRM it lies inside. As shown in Fig. 7, the feasible region of ICR in ASM is determined by the steering limits of the inner wheels. When the turning radius is at its minimum r_{min}, the ICR lies at the intersection of the perpendiculars from the inner wheels at their steering limits. In DMM, the ICR lies at infinity, with its virtual connection to the point of minimum turning radius required to be within the maximum limit δ_{max}.

During an X-to-ZRM switch, if the ICR can move directly and seamlessly from outside to inside when the vehicle is moving, there must be an instant when the ICR lies exactly on the vehicle's contour. At this point, the wheels have already reached 90 degrees, as shown in Fig. 8. As the ICR continues to move inward, the inner wheels would have to turn beyond 90 degrees. However, this is infeasible for most MLAVs due to mechanical steering limits. Similarly, for a 4WID-FWS vehicle, the ICR cannot jump directly from outside to inside when longitudinal velocity exists. Therefore, a more practical and straightforward strategy is to let the vehicle first come to a stop, then steer the wheels to meet the steering constraints of ZRM. This is already the most efficient approach under these conditions.

For switching between ASM and DMM, the ICR remains outside the vehicle contour throughout the process, so no crossing from outside to inside occurs. Thus, there remains potential for improving switching efficiency and smoothness. It should be noted, however, that when the MLAV is in DMM, the ICR is located at infinity. Existing studies have shown that at near-zero yaw rates, the ICR may switch between negative and positive infinity, leading to a singularity [28]. This can cause numerical instability during model-based planning and optimization. To address these issues, this paper proposes a generic transient trajectory planning method for mode switching in MLAVs, which effectively avoids adverse effects caused by singularities.

4.2.1. Trigger conditions for transient planning of mode switching

Consider two adjacent trajectory segments, denoted as Tra_j and Tra_{j+1}, with $N_{\mathrm{tra}_j}+1$ and $N_{\mathrm{tra}_{j+1}}+1$ points, respectively. Let α_1 represent ASM and α_2 represent DMM. Based on the previous analysis, a detailed transient trajectory should be planned when the MLAV switches between ASM and DMM. The transient planner is activated only if the travel direction remains consistent across the mode switching point. The trigger conditions are defined as follows:

$$\begin{cases} P^{v_u}_{\mathrm{tra}_j}(N_{\mathrm{tra}_j}) \cdot P^{v_u}_{\mathrm{tra}_{j+1}}(0) \geq 0; \\ P^{\alpha_1}_{\mathrm{tra}_j}(N_{\mathrm{tra}_j}) + P^{\alpha_2}_{\mathrm{tra}_j}(N_{\mathrm{tra}_j}) \geq 1; \\ P^{\alpha_1}_{\mathrm{tra}_{j+1}}(0) + P^{\alpha_2}_{\mathrm{tra}_j}(0) \geq 1. \end{cases} \tag{35}$$

Here, P_{tra_j} denotes a point in trajectory segment Tra_j, with its specific fields accessed via superscript notation, i.e., $P^{*}_{\mathrm{tra}_j}$. Specifically, $P^{v_u}_{\mathrm{tra}_j}$ denotes the velocity input at that point, as generated by the segmented trajectory planner. Similarly, $P^{\alpha_i}_{\mathrm{tra}_j}$ represents the value of the maneuvering mode variable α_i at that point. Note that if the above condition is not satisfied, the system falls back to the original stop-and-switch strategy.

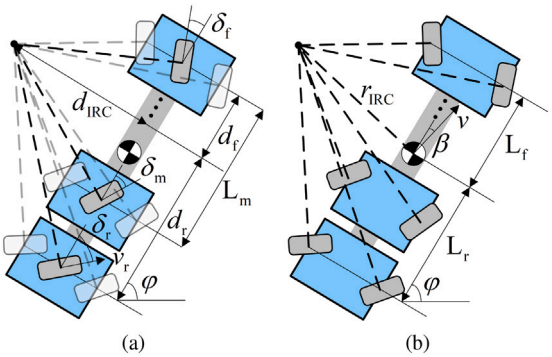


Fig. 9. Modeling approaches of the intermediate mode between ASM and DMM.

4.2.2. Intermediate mode kinematics

The kinematic model for the intermediate mode between ASM and DMM is established. To demonstrate the generality of the proposed modeling method, an nWID-nWIS vehicle (with $n > 2$) is used as an example. Two modeling approaches are then presented.

Method based on virtual single-track control inputs: A virtual single-track vehicle model is constructed, as shown in Fig. 9(a). The distance from the ICR to the longitudinal axis is denoted as d_{ICR}, and the distances to the front and rear axles are denoted as d_f and d_r, respectively. The virtual front steering angle, rear steering angle, and velocity are chosen as control inputs, i.e., $u = \begin{bmatrix}\delta_f & \delta_r & a_r\end{bmatrix}^T$. Clearly, for any given δ_f and δ_r, the ICR is uniquely determined. Using any virtual steering axle as an example, the mapping between its steering angle δ_m, velocity v_m, and the control inputs is derived below.

From the geometric relationship, we have

$$\begin{cases} d_{ICR} = \dfrac{d_f}{\tan\delta_f} = \dfrac{-d_r}{\tan\delta_r} \\ d_f + d_r = L \end{cases} \rightarrow d_{ICR} = \frac{L}{\tan\delta_f - \tan\delta_r}. \tag{36}$$

Since $d_{ICR} = \dfrac{d_f - L_m}{\tan\delta_m}$, substituting into the above yields:

$$\tan\delta_m = \frac{(L - L_m)\tan\delta_f + L_m\tan\delta_r}{L}. \tag{37}$$

Here, L_m denotes the distance from an intermediate axle to the front axle. Similarly, using rigid-body kinematics, the virtual wheel velocity at this axle is derived as follows:

$$v_m = v_r\frac{\cos\delta_r}{\cos\delta_m}. \tag{38}$$

Hence, both the virtual steering angle and velocity of any intermediate axle can be expressed in terms of the control inputs δ_u from the front and rear axles. This implies that the kinematic characteristics of the nWID-nWIS configuration are fundamentally consistent with those of the 4WID-4WIS structure. Furthermore, once all virtual wheel angles and velocities are determined, any two axles can be selected to compute the corresponding real wheel angles and velocities. Denote the steering angles and velocities of the two selected virtual wheels as δ_1, δ_2, v_1, and v_2. The following relation then holds:

$$\begin{cases} \tan\delta_{1l} = \dfrac{2L\tan\delta_1}{2L - B(\tan\delta_1 - \tan\delta_2)},\ v_{1l} = v_1\dfrac{\sin\delta_1}{\sin\delta_{1l}}; \\ \tan\delta_{1r} = \dfrac{2L\tan\delta_1}{2L + B(\tan\delta_1 - \tan\delta_2)},\ v_{1r} = v_1\dfrac{\sin\delta_1}{\sin\delta_{1r}}; \\ \tan\delta_{2l} = \dfrac{2L\tan\delta_2}{2L - B(\tan\delta_1 - \tan\delta_2)},\ v_{2l} = v_2\dfrac{\sin\delta_2}{\sin\delta_{2l}}; \\ \tan\delta_{2r} = \dfrac{2L\tan\delta_2}{2L + B(\tan\delta_1 - \tan\delta_2)},\ v_{2r} = v_2\dfrac{\sin\delta_2}{\sin\delta_{2r}}. \end{cases} \tag{39}$$

Based on the above analysis, an intermediate kinematic model for the nWID-nWIS MLAV can be derived using a coordinate system centered at the rear axle:

$$\begin{cases} \dot{x}_r = v_r\cos\left(\varphi + \delta_r\right), \\ \dot{y}_r = v_r\sin\left(\varphi + \delta_r\right), \\ \dot{\varphi} = \dfrac{v_r\cos\delta_r\left(\tan\delta_f - \tan\delta_r\right)}{L}, \\ \dot{v}_r = a_r. \end{cases} \tag{40}$$

Table 2
Control input variables for the transient planning of maneuvering mode switching.

Chassis configuration	Equivalent control inputs of intermediate mode kinematics
nWID-nWIS	$[\delta_f,\delta_r,a_r]^T$ or $[\beta,\theta,a]^T$
nWID-nWS	$[\delta_f,\delta_r,a_r]^T$ or $[\beta,\theta,a]^T$
nWD-nWS	$[\delta_f,\delta_r,a_r]^T$ or $[\beta,\theta,a]^T$

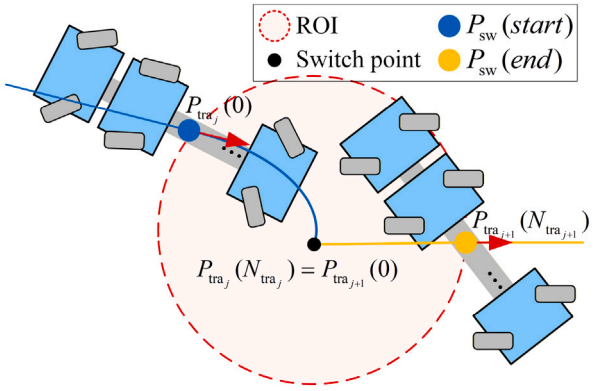


Fig. 10. Illustration of the ROI determination method.

Method based on the ICR: As shown in Fig. 9(b), the vehicle's velocity at the CoG is denoted as v, and the angle between this velocity and the longitudinal axis is the sideslip angle β. The turning radius r_{ICR} is defined as the distance from the CoG to the ICR. An auxiliary angle $\tan\theta = \frac{1}{r_{ICR}}$ is introduced [30]. Note that θ approaches zero when the MLAV is in diagonal motion mode. The sideslip angle β, the auxiliary angle θ, and the acceleration a of the CoG are selected as control inputs, i.e., $u = \begin{bmatrix}\beta & \theta & a\end{bmatrix}^T$. The following kinematic relations are established:

$$\begin{cases} \dot{x} = v\cos(\varphi+\beta), \\ \dot{y} = v\sin(\varphi+\beta), \\ \dot{\varphi} = v\tan\theta, \\ \dot{v} = a. \end{cases} \tag{41}$$

To maintain consistency with the steering angle used as the control input in the path optimization stage, a mapping from the steering angles to the sideslip angle β and the auxiliary angle θ must be established. Using the virtual front and rear steering angles δ_f and δ_r defined in Method 1, the following relation is readily available:

$$\begin{cases} \tan\delta_f = \dfrac{L_f\tan\theta + \sin\beta}{\cos\beta}, \\ \tan\delta_r = -\dfrac{L_r\tan\theta - \sin\beta}{\cos\beta}. \end{cases} \tag{42}$$

The steering angle δ_m of any intermediate axle can then be derived from δ_f and δ_r according to (37). Then the actual wheel speeds and steering angles can subsequently be calculated using (39).

Both modeling approaches ensure that the control inputs and system states vary continuously without numerical issues, even when the ICR switches between negative and positive infinity. As a result, both methods avoid the singularity issue described in [28]. The kinematic equations above can be compactly expressed as:

$$\dot{\xi} = f_{sw}(\xi,u). \tag{43}$$

Remark 4. The proposed transient planning for mode switching is applicable to any multi-axle vehicle as long as each axle has steering capability. Such vehicles can execute at least Ackermann steering and diagonal motion modes. The selection of control input variables for other chassis configurations is detailed in Table 2.

4.2.3. Definition of the virtual ROI

A virtual ROI is established for transient planning of mode switching, as shown in Fig. 10. The mode switching point is defined as the last point $P_{tra_j}(N_{tra_j})$ of segment Tra_j or the first point $P_{tra_{j+1}}(0)$ of segment Tra_{j+1}. A circular region of radius r_{ROI} is constructed centered at this switching point, where the radius equals the arc length along the trajectory from the switching point to both P_{sw}(start)

and $P_{\text{sw}}(\text{end})$:

$$r_{\text{ROI}} = \int_{P_{\text{sw}}(\text{start})}^{P_{\text{tra}_j}(N_{\text{tra}_j})} \mathrm{d}s = \int_{P_{\text{tra}_{j+1}}(0)}^{P_{\text{sw}}(\text{end})} \mathrm{d}s. \tag{44}$$

where $\mathrm{d}s$ denotes the differential arc length along the trajectory. For discrete trajectory representations, r_{ROI} is computed by summing the Euclidean distances between consecutive points along the trajectory segments. The intersections of the ROI boundary with Tra_j and Tra_{j+1} are denoted as $P_{\text{sw}}(\text{start})$ and $P_{\text{sw}}(\text{end})$, representing the start and end of the mode switching. To facilitate subsequent formulation, quantities such as pose and control inputs at these points are accessed via superscript notation, e.g., P_{sw}^{ξ}.

4.2.4. Transient planning problem of mode switching

The general form of the transient planning problem is given below:

$$\min J_3 \tag{45}$$

$$\text{s.t. } \xi(k+1) = \xi(k) + \frac{T}{N_{\text{ROI}}} f_{\text{sw}}\big(\xi(k), u(k)\big),\ k \in I_0^{N_{\text{ROI}}}; \tag{45a}$$

$$\xi(0) = P_{\text{sw}}^{\xi}(\text{start}),\ \xi(N_{\text{ROI}}) = P_{\text{sw}}^{\xi}(\text{end}); \tag{45b}$$

$$u(0) = P_{\text{sw}}^{u}(\text{start}),\ u(N_{\text{ROI}}-1) = P_{\text{sw}}^{u}(\text{end}); \tag{45c}$$

$$C_i^{\text{disc}}(k) \in STC_i(k),\ i \in I_{\text{feature}},\ k \in I_0^{N_{\text{ROI}}}; \tag{45d}$$

$$\delta_{\min} \le \left\| \psi_{\delta}^{-1}\big(\delta_u(k)\big) \right\|_{\infty} \le \delta_{\max},\ k \in I_0^{N_{\text{ROI}}-1}. \tag{45e}$$

Here, N_{ROI} denotes the number of trajectory points within the ROI. Constraint (45a) represents the intermediate kinematic constraints between vehicle maneuvering modes. Constraints (45b) and (45c) specify the pose and control information at the start and end of the mode switching process. Constraint (45d) enforces collision avoidance, while constraint (45e) ensures that all steering angles remain within mechanical limits. The cost function J_3 can be customized as needed. For example, to optimize switching efficiency, smoothness, and stability, the following cost function can be used:

$$J_3 = \eta_{\text{yaw}} \sum_{k=0}^{N_{\text{ROI}}-1} \omega^2(k) + \eta_{\text{lateral}} \sum_{k=0}^{N_{\text{ROI}}-1} \left[v_y(k+1) - v_y(k) \right]^2 + \eta'_{\text{time}} T^2, \tag{46}$$

where η_{yaw}, η_{lateral}, and η'_{time} represent the associated weight coefficients. The first term penalizes the yaw rate, the second term penalizes the rate of change of the lateral velocity v_y at the CoG, and the third term on the right side penalizes the total switching time. Thus, larger values of η_{yaw}, η_{lateral}, and η'_{time} correspond to stronger emphasis on yaw stability, lateral velocity variation, and mode-switching efficiency, respectively. For example, under relatively high-speed conditions, η_{yaw} and η_{lateral} are increased to prioritize stability and smoothness at the expense of some switching efficiency. If the first method is used to model the transitional kinematics for the nWID-nWIS vehicle, we have:

$$\begin{cases} \omega(k) = \dfrac{v_{\text{r}}(k)\cos\delta_{\text{r}}(k)\left(\tan\delta_{\text{f}}(k) - \tan\delta_{\text{r}}(k)\right)}{\text{L}}, \\ v_y = v_{\text{r}}(k)\sin\delta_{\text{r}}(k) + \omega(k)\text{L}_{\text{r}}. \end{cases} \tag{47}$$

If the second method is used, the function becomes:

$$\begin{cases} \omega(k) = v(k)\tan\theta(k), \\ v_y(k) = v(k)\sin\beta(k). \end{cases} \tag{48}$$

By solving the above optimization problem, a smooth switching trajectory $\mathcal{T}_{\text{sw},j}$ within the ROI can be obtained. This trajectory is then used to directly replace the original one, resulting in a complete and suitable trajectory $\mathcal{T}$ for the MLAV's multi-mode motion.

4.3. Implementation workflow and complexity analysis

4.3.1. Overall workflow

Algorithm 1 describes the implementation workflow of the proposed framework. The algorithm takes the start state ξ_{start}, the goal state ξ_{goal}, and the environment map $\mathcal{M}$ as inputs. The final output is a feasible spatio-temporal trajectory $\mathcal{T}$.

The first layer focuses on spatial planning through four main steps. First, the CoarsePathSearch() function determines an initial path $\mathcal{P}_{\text{coarse}}$ and a sequence of maneuvering modes $\{M_i\}$. Second, the ConstructSafeCorridor() function generates safe corridors to provide collision-avoidance constraints. Third, the UpdateBounds() function calculates the bounds for the big-M method in (16). Finally, the PathModeOptimization() function optimizes the path geometry and modes together by solving (30).

The second layer handles spatio-temporal planning. The optimized spatial path is first divided into several segments based on mode consistency. Each segment seg_j is then processed by the PiecewiseTrajectoryGen() function, which solves the optimization problem (33) to generate a local trajectory. If a mode switching is required between adjacent segments, the SwitchingPlan() function is executed by solving the optimization problem (45). Finally, the SynthesizeSegments() function assembles all segments into the final trajectory $\mathcal{T}$ for output.

Algorithm 1: Generic spatio-temporal decoupling planning framework.

Input : Start state ξ_{start}, Goal state ξ_{goal}, Environment map $\mathcal{M}$
Output: Trajectory $\mathcal{T}$

/* Layer 1: Spatial Planning */
$\mathcal{P}_{\text{coarse}}, \{M_i\} \leftarrow \text{CoarsePathSearch}(\xi_{\text{start}}, \xi_{\text{goal}}, \mathcal{M})$; // Multi-mode path search
$\{STC_i\} \leftarrow \text{ConstructSafeCorridor}(\mathcal{P}_{\text{coarse}}, \mathcal{M})$; // Generate STC constraints
$m_\xi, M_\xi \leftarrow \text{UpdateBounds}(\boldsymbol{\beta}, \mathcal{P}_{\text{coarse}})$; // Update bounds in constraints (16)
$\mathcal{P}^*, \{M_i^*\} \leftarrow \text{PathModeOptimization}(\mathcal{P}_{\text{coarse}}, \{STC_i\})$; // Co-optimization of path and modes
/* Layer 2: Spatio-temporal Planning */
Divide $\mathcal{P}^*$ into $N_{\text{sw}}+1$ segments $\{\text{seg}_j\}_{j=1}^{N_{\text{sw}}}$ based on mode consistency;
foreach *segment* seg_j **do**
  $\text{Tra}_j \leftarrow \text{PiecewiseTrajectoryGen}(\text{seg}_j)$; // Independent segment optimization
for $j \leftarrow 1$ **to** N_{sw} **do**
  if *trigger condition (35) is activated between* Tra_j *and* Tra_{j+1} **then**
    $\mathcal{T}_{\text{sw},j} \leftarrow \text{SwitchingPlan}(\text{Tra}_j, \text{Tra}_{j+1})$; // Switching optimization
$\mathcal{T} \leftarrow \text{SynthesizeSegments}(\{\mathcal{T}_{\text{sw},j}\})$; // Final trajectory assembly
return $\mathcal{T}$;

4.3.2. Computational complexity analysis

The computational performance of the proposed hierarchical framework is analyzed by examining each module. This analysis focuses on how the time complexity scales with the planning horizon, the number of modes, and the environment.

First, for the coarse search, the worst-case complexity is $O\left(\left(\sum_{i=1}^{n_{\text{m}}} b_{\text{exp},M_i}\right)^{d_{\text{srch}}}\right)$. Here, b_{exp,M_i} is the number of child nodes expanded under mode M_i, and d_{srch} is the search depth. In practice, the relaxed termination condition and inflated heuristic cost reduce the search time.

Second, for the path optimization layer, the number of integer variables per step is $n_I = n_{\text{m}}+1$. The number of continuous variables per step is $n_C = (n_{\text{m}}+1)(n_\xi + n_u)$, where n_ξ and n_u are the dimensions of the state vector and control input vector, respectively. Let $T_{\text{cont}}(Nn_C)$ represent the cost of solving the nonlinear sub-problems on each branch. The resulting complexity is $O(2^{Nn_I} \cdot T_{\text{cont}}(Nn_C))$. The complexity increases exponentially as the planning horizon N grows. To reduce this burden, confidence level $\boldsymbol{\beta}$ is used to tighten the big-M bounds, and cutting planes are employed to prune redundant branches.

Third, the spatio-temporal layer is computationally cheaper because the trajectory is optimized segment by segment. With the interior-point method, the complexity of the piecewise formulation is $\sum_{j=1}^{N_{\text{sw}}+1} O((N_{\text{seg}_j}(n_\xi + n_u))^3)$. This is much lower than the complexity of a non-segmented approach, which would be $O\left((\sum_{j=1}^{N_{\text{sw}}+1} N_{\text{seg}_j}(n_\xi + n_u))^3\right)$.

Finally, for the transient planning for mode switching, if N_{trig_sw} switches are triggered and each ROI contains N_{ROI} points, the time complexity is $N_{trig_sw} \cdot O((N_{\text{ROI}}(n_{\xi_\text{int}} + n_{u_\text{int}}))^3)$. Small ROI radii are utilized to ensure that the framework remains efficient and suitable for near real-time applications.

4.3.3. Practical applicability

The proposed framework demonstrates high flexibility for both offline planning and near real-time operations. To clarify its practical applicability, we explicitly distinguish the conditions under which the method operates reliably from those requiring full replanning.

The framework operates reliably in two main scenarios. First, in static environments, the offline spatial layer generates high-quality reference trajectories that remain fully valid. Second, in mildly dynamic environments, the method remains reliable if a dynamic obstacle only intrudes on the middle of a planned trajectory segment. Because the spatial layer already defines the mode sequence and switching points, the spatio-temporal layer can perform fast local replanning. It adjusts the intermediate positions and velocities to avoid the obstacle without changing the segment's endpoints. This near real-time local replanning ensures safe navigation while efficiently preserving the overall multi-mode strategy.

Full replanning becomes necessary when a dynamic obstacle persistently occupies a mode-switching point. The current spatio-temporal layer relies on these fixed endpoints to smoothly connect adjacent segments. If an obstacle blocks a switching point, local trajectory adjustments cannot resolve the conflict. In this situation, the system must trigger a full replanning from the spatial layer to recalculate the geometric path and relocate the switching points. Because this spatial search requires more computational time, the framework currently has limitations in highly dynamic environments where switching points are frequently obstructed. Addressing high-frequency feedback for these specific scenarios will be a focus of our future research.

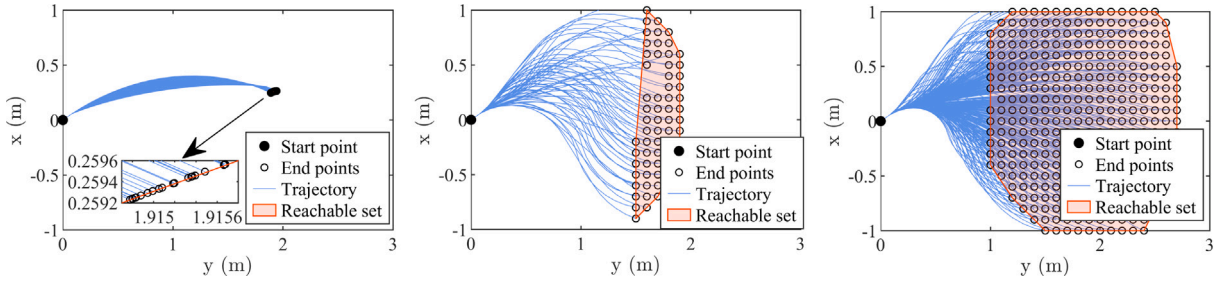
Table 3
Vehicle model Specifications for simulation.

Symbol	Description	Value
L	Wheelbase [m]	2.91
B	Track width [m]	1.675
L_f	Distance from CoG to front axle [m]	1.25
L_r	Distance from CoG to rear axle [m]	1.66
L_{fh}	Front overhang [m]	0.5
L_{rh}	Rear overhang [m]	0.5

Table 4
Scenario setup for feasibility verification.

Algorithm	Initial pose [m],[deg]	Initial wheel angle [deg][1]	Final pose [m],[deg]	Final wheel angle [deg]	Switching duration [s]	Speed [m/s]	Wheel angle profiles
Baseline	(0, 0), 0	0, 0, 30.23, 41.25	*[2]	−30.98, −42.53, 0, 0	1.9	1	*
Ours (constant speed)	(0, 0), 0	0, 0, 30.23, 41.25	*	−30.98, −42.53, 0, 0	1.9	1	*
Ours (varying speed)	(0, 0), 0	0, 0, 30.23, 41.25	*	−30.98, −42.53, 0, 0	1.9	0.5–1.5	*

[1] Values correspond to wheel angles in the order: front-left, front-right, rear-left, rear-right.
[2] To be optimized.



(a) Baseline (b) Ours (constant speed) (c) Ours (varying speed)

Fig. 11. Comparison of trajectory clusters and reachable sets under conditions in Table 4. Blue lines show clusters of feasible trajectories, and the red region depicts corresponding reachable sets of end points. Specifically, the reachable set of the baseline is nearly a line.

5. Simulation results and discussions

Simulation studies for motion planning are conducted using a 4WID-4WIS vehicle. The vehicle parameters are listed in Table 3. The simulation environment is implemented in Python and executed on a computer equipped with an Intel i5-11260H CPU @ 2.60 GHz. The MIP problem presented in Section 3.3 is solved using the Gurobi solver [31], while the nonlinear programming problem in Section 4 is solved with IPOPT [32].

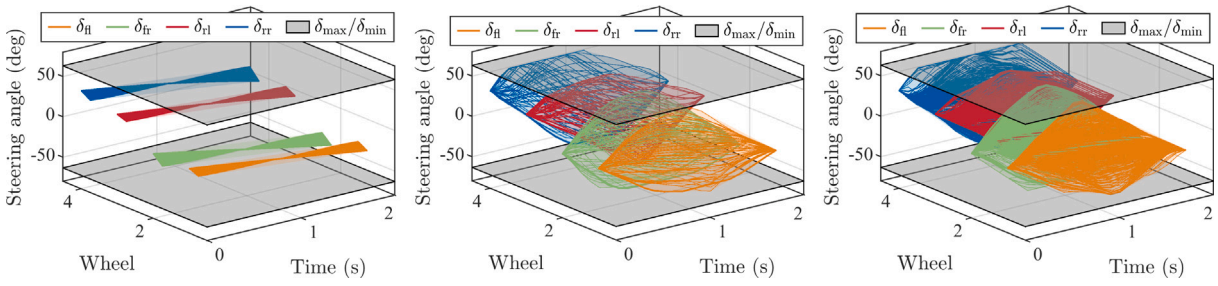
5.1. Performance evaluation of transient trajectory planning for mode switching

This subsection evaluates the proposed transient trajectory planning method for mode switching in terms of feasibility, smoothness, and stability. A comparison is made with a baseline method that uses a constant turning radius and generates steering angle profiles via B-spline interpolation [24]. The baseline method first computes the target steering angles for all wheels at the end of the transition based on the constant-radius assumption. Then, it generates the steering angle profiles over time by fitting sampled B-spline control points. Finally, the corresponding spatio-temporal trajectories are derived from these profiles.

5.1.1. Feasibility

To ensure a fair comparison of feasible regions, the simulation uses the following uniform settings: the initial pose of the vehicle is fixed, a constant longitudinal speed is maintained during the switching interval, and the steering angles at both the start and end of the switching are the same. Under these common constraints, both the baseline method and the proposed method are used to generate their respective trajectory clusters. The feasible regions formed by the two trajectory clusters are then compared. Detailed simulation parameters are listed in Table 4. Fig. 11 shows the vehicle trajectory clusters and reachable sets, while Fig. 12 presents the optimizable range of the steering angles.

The simulation results show that under the same conditions, the trajectory clusters generated by the proposed method cover a significantly larger feasible region than those of the baseline. For the baseline method, the reachable set is almost a single curve, indicating that the terminal position and the path are almost completely predetermined once the initial steering angle is fixed, with only minor variations possible. In contrast, the proposed method clearly offers greater diversity in trajectory generation and a wider feasible region. Furthermore, when the constant longitudinal speed constraint is relaxed, the feasible region of the proposed method

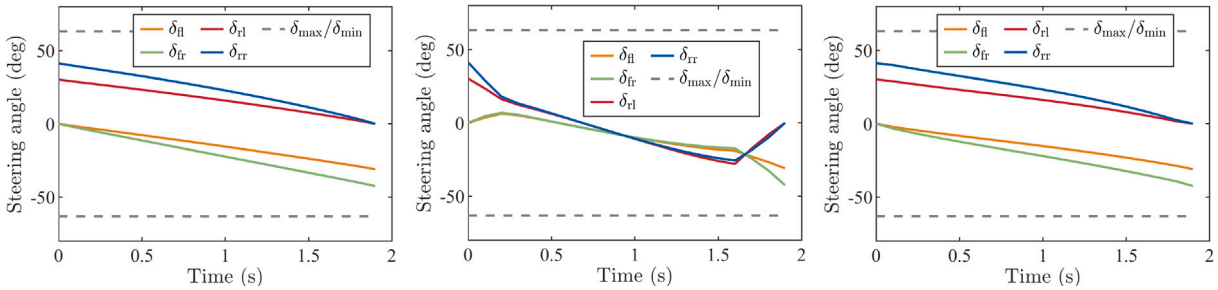


(a) Baseline (b) Ours (constant speed) (c) Ours (varying speed)

Fig. 12. Comparison of feasible steering angle profile clusters given the conditions in Table 4. Wheel axis values 1–4 correspond to front-left, front-right, rear-left, and rear-right wheels, respectively.

Table 5
Scenario setup for smoothness and stability verification.

Algorithm	Initial pose [m],[deg]	Initial wheel angle [deg]	Final pose [m],[deg]	Final wheel angle [deg]	Switching duration [s]	Speed [m/s]	Wheel angle profiles
Baseline	(0, 0), 0	0, 0, 30.23, 41.25	(1.89, 0.25), 25.13	−30.98, −42.53, 0, 0	1.9	1	*
Ours (cluster)	(0, 0), 0	0, 0, 30.23, 41.25	*	−30.98, −42.53, 0, 0	1.9	1	*
Ours (point-to-point)	(0, 0), 0	0, 0, 30.23, 41.25	(1.89, 0.25), 25.13	−30.98, −42.53, 0, 0	1.9	1	*



(a) Baseline (b) Ours (cluster) (c) Ours (point-to-point)

Fig. 13. Comparison of wheel steering angle profiles obtained with the optimal cost under the conditions in Table 5.

expands, whereas the baseline method lacks the ability to plan velocity profiles. Similarly, as shown in Fig. 12, the proposed method provides a much broader optimizable range for steering angle profiles during switching transients.

It should be noted that the above conclusions are derived from simulations where the initial and final steering angles are fixed. In practical applications of the proposed method, the steering angle at the terminal moment is also an optimized variable, which would further improve both feasibility and optimality. This simulation compares the proposed algorithm with the baseline under relatively fair conditions, and therefore intentionally restricts part of the feasibility and optimality of the proposed approach.

5.1.2. Smoothness and stability

From each resulting region of trajectory clusters, an optimal switching profile is selected for each method based on predefined performance metrics, such as the integral of lateral velocity variation or the accumulated yaw rate. The cost function is defined as follows:

$$J_3' = \eta_{\text{yaw}} \sum_{k=0}^{N_{\text{ROI}}-1} \omega^2(k) + \eta_{\text{lateral}} \sum_{k=0}^{N_{\text{ROI}}-1} \left[v_y(k+1) - v_y(k)\right]^2, \tag{49}$$

where $\eta_{\text{yaw}} = 1$ and $\eta_{\text{lateral}} = 5$ weight the yaw rate and lateral velocity variation terms, respectively. To further validate optimality, the terminal position and heading of the optimal trajectory from the baseline method are used as endpoint constraints. Under these constraints, the proposed method is applied to search for a new transient switching trajectory, denoted as “ours (point-to-point)”. Detailed simulation settings are provided in Table 5.

Figs. 13 and 14 illustrate the evolution of the vehicle’s steering angle and pose, respectively. The baseline method produces an arc-like trajectory at the CoG, which results from the constant steering radius principle. When the terminal pose is not constrained, the proposed method allows more flexible motion, with the vehicle exhibiting nearly diagonal movement during mode switching to reduce yaw motion. When the terminal pose is constrained to match that of the baseline’s optimal trajectory, the resulting path closely aligns with the baseline solution in both spatial shape and temporal behavior. This confirms that the trajectory generated by the constant-radius method constitutes one specific feasible solution within the solution set of the proposed approach. Fig. 15

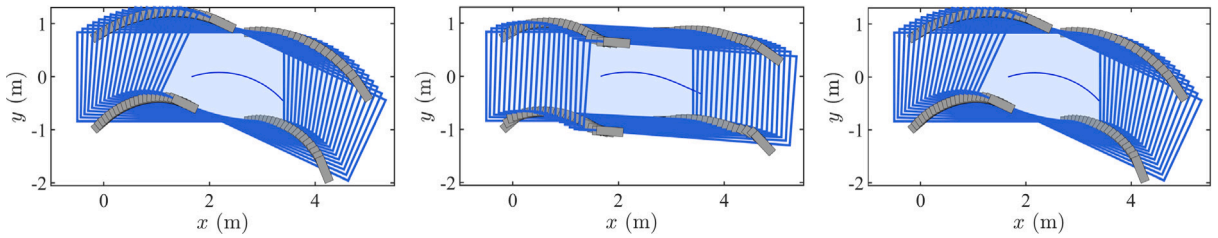


Fig. 14. Comparison of pose evolution obtained with the optimal cost under the conditions in Table 5.

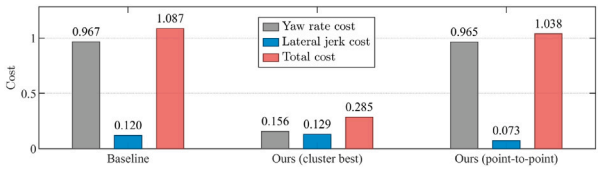


Fig. 15. Statistical results of the optimal cost.

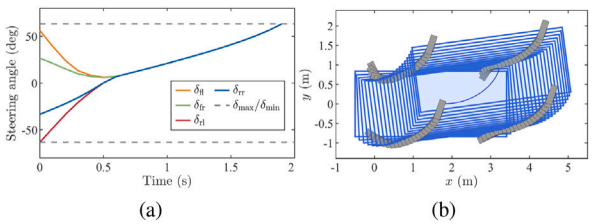


Fig. 16. Switching results from ASM to DMM of proposed method.

compares the costs of the three methods, including yaw rate cost, lateral velocity variation cost, and total cost. The results indicate that the proposed method significantly reduces lateral velocity variation without substantially increasing yaw rate, leading to a lower total cost and better overall smoothness and stability than the baseline. Even under terminal pose constraints, the proposed method still achieves slightly better performance metrics than the baseline, demonstrating that the baseline solution is included in the feasible set of the proposed method and highlighting the proposed method's advantages in diversity and optimality of solutions.

In summary, under the same conditions, the proposed method outperforms the baseline in feasibility, smoothness, and stability. These results demonstrate its clear advantage in generating higher-quality and more flexible switching trajectories.

Remark 5. The proposed transient planning method effectively handles switches involving DMM. As shown in Fig. 16, the vehicle begins in a maximum left turn ASM, with the inner left rear wheel reaching its steering limit. The target state is a left-forward DMM, also at the steering limit. Using the proposed method, the vehicle switches smoothly from the initial ASM to the desired DMM while satisfying the mechanical limits of all steering wheels.

5.2. Case studies of the generic motion planning framework

This section evaluates the effectiveness and advantages of the proposed generic motion planning method for multi-mode vehicles through three representative scenarios:

Scenario 1: Parallel parking with an obstructing obstacle above the parking space.

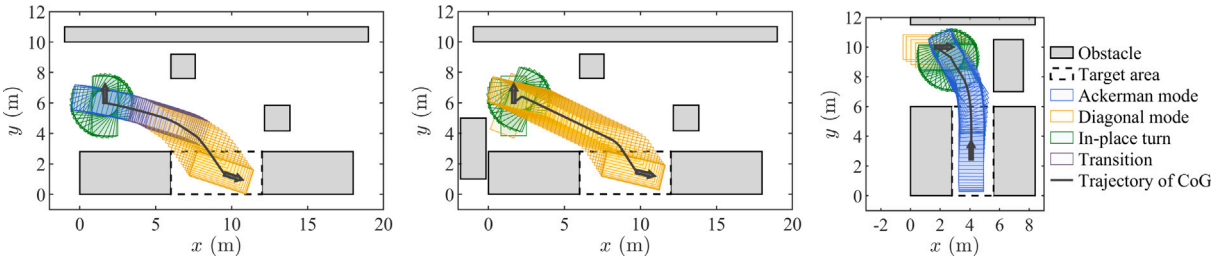
Scenario 2: Extending Scenario 1, an illegally parked vehicle is placed near the initial position.

Scenario 3: Vertical parking in a narrow passage with static obstructing obstacles.

1) Comparison with the classical single-mode planning method

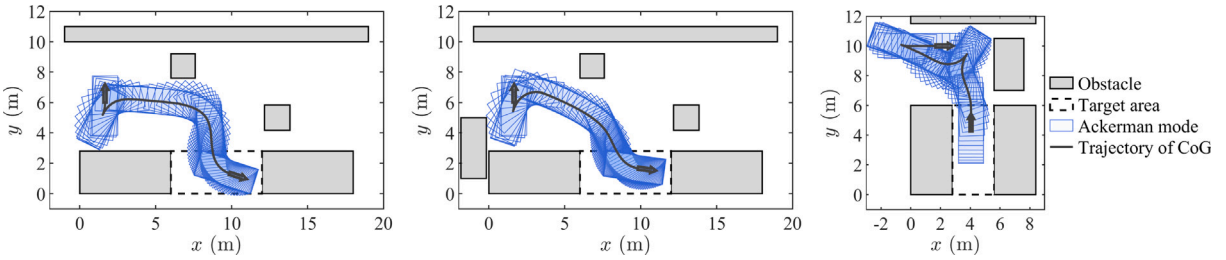
To demonstrate the advantages of multi-mode motion planning in complex environments, a classical single-mode planning method is introduced (denoted as Baseline 1) for comparison [33]. This baseline method uses Hybrid A* to generate an initial path, combines STC to build obstacle avoidance constraints, and employs nonlinear programming for trajectory optimization. To ensure a fair comparison, both methods aim to safely reach the designated parking area without imposing additional constraints on the final parking pose.

The trajectories resulting from both methods are shown in Figs. 17 and 18. Both approaches successfully complete the parking task without collisions. However, since the baseline method only supports ASM, it relies on frequent forward and backward switching in narrow spaces, resulting in a curved and inefficient path. In contrast, the proposed method autonomously selects appropriate motion modes, generating a more direct and efficient trajectory without noticeable cusps.



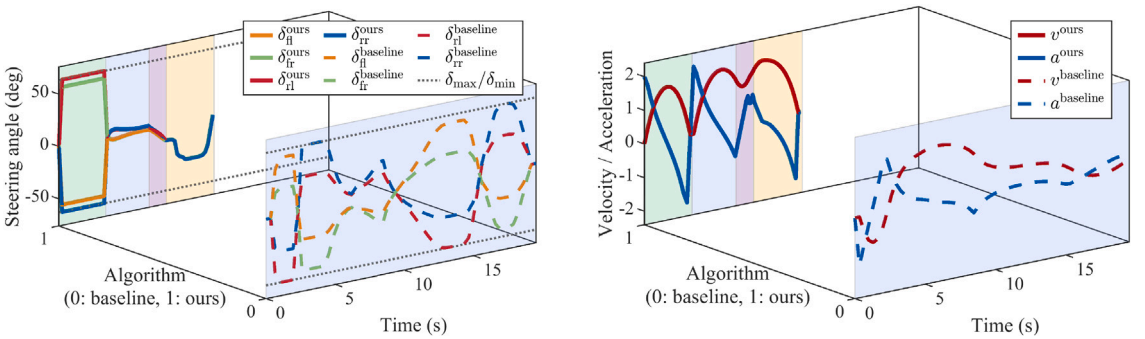
(a) Scenario 1 (b) Scenario 2 (c) Scenario 3

Fig. 17. Pose evolution under the proposed method.



(a) Scenario 1 (b) Scenario 2 (c) Scenario 3

Fig. 18. Pose evolution under Baseline 1.



(a) Steering angle profiles (b) Speed and acceleration profiles

Fig. 19. Vehicle state and control profiles under scenario 1. The green, blue, purple, and yellow regions represent the ZRM, ASM, switching transient, and DMM, respectively.

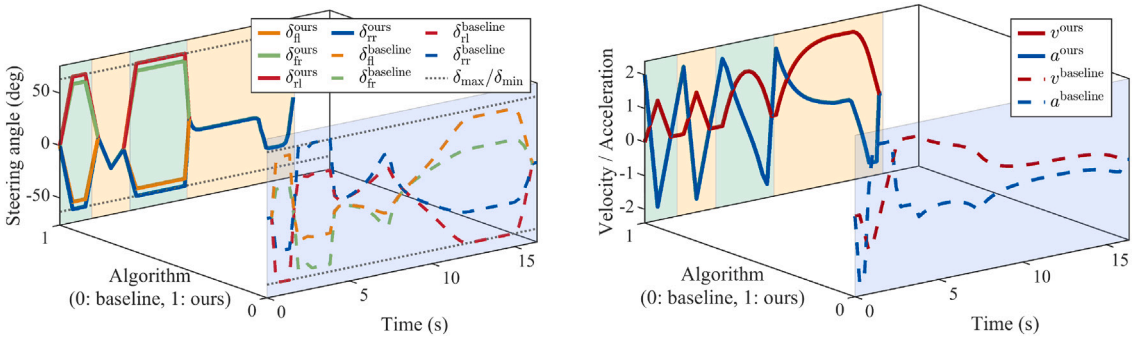
To enable a quantitative comparison, Figs. 19-21 show the steering angle and velocity profiles of both methods across the three scenarios. The results indicate that both methods satisfy steering angle constraints, but the total travel time of Baseline 1 is consistently longer than that of the proposed method in all scenarios. This suggests that single-mode planning is less capable of utilizing the vehicle's motion capabilities. It is worth noting that although mode switches involving point turns require the vehicle to stop briefly, which reduces both velocity and steering angle to zero, reasonable multi-mode decision-making and switching still significantly reduce the overall travel time. In Scenario 1, the vehicle achieves a smooth switch from ASM to DMM without stopping, further improving operational efficiency.

The computational results in Table 6 show that the proposed method is faster than Baseline 1 in all scenarios. The modified A* provides efficient multi-mode guidance, and the segmented optimization reduces the cost of spatio-temporal trajectory generation.

2) Comparison with methods that separate path planning and mode decisions

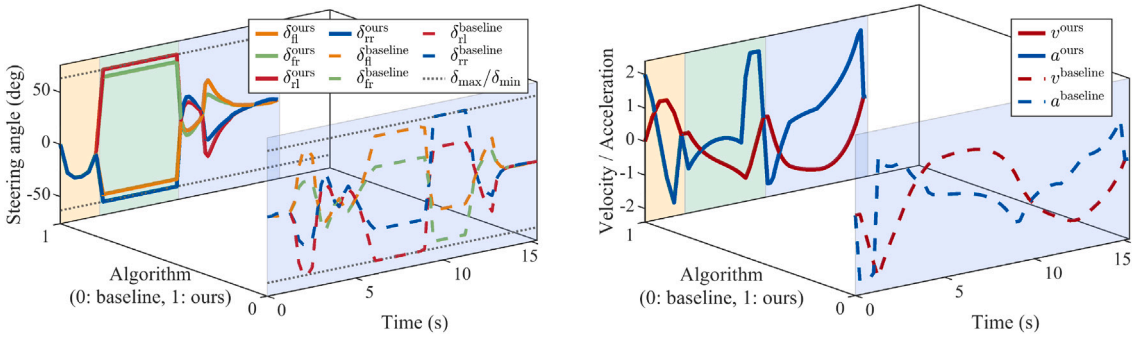
To further demonstrate the benefits of co-optimizing the geometric path and maneuvering mode sequence, two additional baselines are introduced. Baseline 2 first generates a coarse path using standard two-dimensional A*, then assigns maneuvering modes through a rule-based fuzzy selector [22], and finally performs piecewise trajectory optimization. Baseline 3 follows the same procedure, except that its coarse path is generated by the modified A* in Section 3.1.

The fuzzy selector is constructed according to the mode definitions and the basic framework in [22]. It should be noted that the maneuvering mode taxonomy in [22] is slightly different from that adopted in this paper. Specifically, independent wheel steering (IWS) is equivalent to DMM. The Ackermann steering mode is further divided into AS_drive and AS_spin, which correspond to ASM and ZRM, respectively. In addition, differential steering (DS) is considered in [22] and is therefore retained in the constructed baseline.



(a) Steering angle profiles (b) Speed and acceleration profiles

Fig. 20. Vehicle state and control profiles under scenario 2. The green, blue, and yellow regions represent the ZRM, ASM, and DMM, respectively.



(a) Steering angle profiles (b) Speed and acceleration profiles

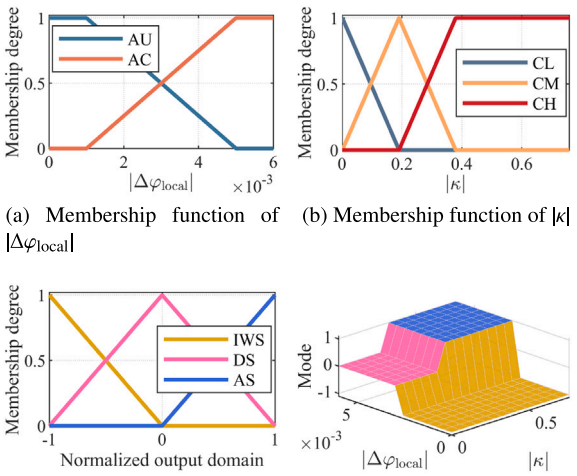
Fig. 21. Vehicle state and control profiles under scenario 3. The green, blue, and yellow regions represent the ZRM, ASM, and DMM, respectively.

Table 6
Computation time for different scenarios (unit: s).

Scenario	Proposed method [s]					Baseline 1 [s]		
	Improved A*	Path optimization	Trajectory generation	Transient planning for switching	Total	Hybrid A*	Trajectory optimization	Total
1	0.35	2.87	0.07	0.04	3.32	7.02	4.36	11.38
2	0.39	3.30	0.03	—	3.72	5.37	3.48	8.85
3	0.17	2.31	0.08	—	2.56	3.67	4.55	8.22

The fuzzy selector takes two inputs, namely the absolute path curvature $|\kappa|$ and the absolute local heading variation $|\Delta\varphi_{\text{local}}|$, both defined on nonnegative domains. For $|\Delta\varphi_{\text{local}}|$, two fuzzy sets are defined, namely attitude unchanged (AU) and attitude changeable (AC). For $|\kappa|$, three fuzzy sets are defined, namely curvature low (CL), curvature medium (CM), and curvature high (CH). The output is a normalized mode variable $z \in \{-1,0,1\}$, where $z = -1$, $z = 0$, and $z = 1$ correspond to IWS, DS, and AS, respectively. The membership functions of the input and output are shown in Figs. 22(a)-(c). If the local heading remains unchanged, IWS is selected. If the heading changes, the selector distinguishes between DS and AS according to the curvature level. DS is preferred in the CL and CM regions, whereas AS is activated in the CH region because DS may induce more severe tire wear under large curvature. The AS spin mode is not determined by the fuzzy inference system. Instead, it is identified separately by a spin detector before fuzzy mode selection. Based on the above "IF-THEN" logic, Fig. 22(d) illustrates the relationship between the input and output.

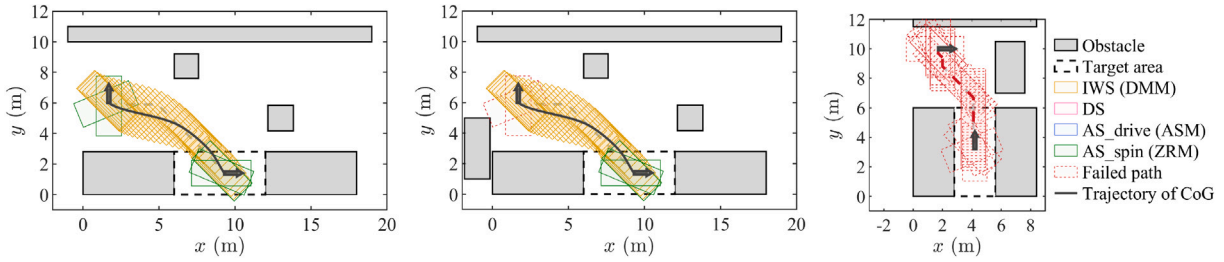
Simulation results are shown in Figs. 23 and 24. It can be observed that neither baseline successfully handles all scenarios. For Baseline 2, the coarse path search ignores the vehicle's heading and multi-mode characteristics. In relatively open environments like Scenario 1, this approach remains effective. However, in narrow spaces with more obstacles, selecting maneuvering modes based only on curvature and heading change becomes inadequate, as it neglects the coupling effects of the surrounding environment. The resulting low-quality coarse path restricts the optimization freedom in later steps, thus leading to infeasibility. This is evident in Fig. 23(b) at the initial segment and throughout the trajectory in Fig. 23(c). For Baseline 3, the coarse path search incorporates multi-mode characteristics, leading to more reasonable path generation. In Fig. 24(b), for instance, the vehicle actively avoids the obstacle near its initial position, demonstrating the effectiveness of the proposed modified A* method. However, the mode selection still lacks consideration of kinematic feasibility, resulting in subsequent infeasible segments. In Fig. 24(c), the vehicle successfully reaches the target by using AS_drive (ASM) to bypass the left-side obstacle. Yet compared to the proposed method in Fig. 17(c), its mode choice is clearly suboptimal. These results demonstrate the necessity of co-optimizing the geometric path and maneuvering mode sequence.



(a) Membership function of $|\Delta\varphi_{\text{local}}|$ (b) Membership function of $|\kappa|$

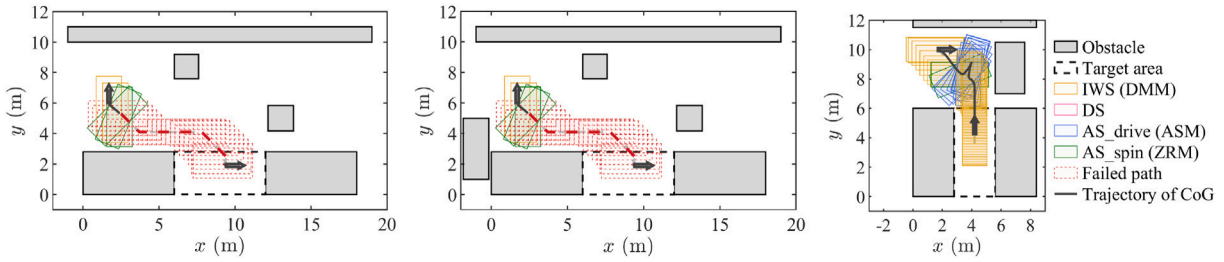
(c) Membership function of maneuvering mode (d) Relationship between the input and output

Fig. 22. Fuzzy selector for maneuvering modes in baseline 2 and baseline 3.



(a) Scenario 1 (b) Scenario 2 (c) Scenario 3

Fig. 23. Pose evolution under baseline 2. The red box labeled "Failed path" indicates a failure in segmented trajectory optimization. In these cases, only the coarse path is displayed.



(a) Scenario 1 (b) Scenario 2 (c) Scenario 3

Fig. 24. Pose evolution under baseline 3. The red box labeled "Failed path" indicates a failure in segmented trajectory optimization. In these cases, only the coarse path is displayed.

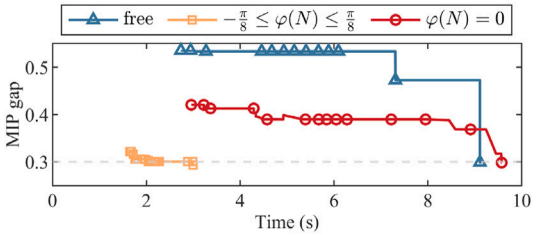
Such co-optimization avoids unnecessary detours and infeasibility caused by incomplete consideration of multi-mode characteristics and the surrounding environment.

3) Discussion on the impact of terminal constraints on solution efficiency

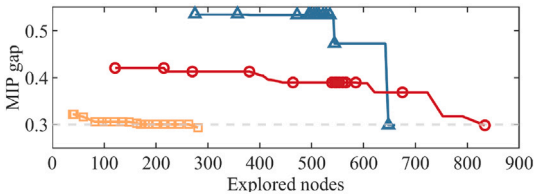
To further examine the effect of terminal constraints on the MIP, we compared three terminal-heading settings for Scenario 1: no additional constraint on $\varphi(N)$, a bounded constraint $\varphi(N) \in [-\pi/8, \pi/8]$, and an exact equality constraint $\varphi(N) = 0$. Fig. 25 shows the evolution of the relative MIP gap with respect to solver runtime and explored nodes, respectively. The relative MIP gap is computed as

$$\text{gap} = \frac{|z_{\text{inc}} - z_{\text{bnd}}|}{|z_{\text{inc}}|}, \tag{50}$$

where z_{inc} denotes the objective value of the current incumbent feasible solution and z_{bnd} denotes the current best bound returned by the branch-and-bound procedure. Since the present formulation is a minimization problem, z_{bnd} corresponds to the best lower bound.



(a) MIP gap over time



(b) MIP gap over explored nodes

Fig. 25. Effect of terminal constraint settings on MIP gap illustrated by Scenario 1.

Table 7
Specifications of the 4WID-4WS prototype vehicle.

Symbol	Description	Value
L	Wheelbase [m]	0.89
B	Track width [m]	0.82
L_f	Distance from CoG to front axle [m]	0.445
L_r	Distance from CoG to rear axle [m]	0.445
L_{fh}	Front overhang [m]	0.165
L_{rh}	Rear overhang [m]	0.29

Therefore, a smaller gap indicates that the incumbent solution is closer to proven optimality. It can be observed that the bounded terminal constraint $\varphi(N) \in [-\pi/8,\pi/8]$ yields the best overall performance. In particular, it leads to the shortest time to obtain the first feasible solution and also the shortest time to reduce the relative MIP gap to 0.3.

This behavior can be explained from the viewpoint of branch-and-bound. When no terminal-heading constraint is imposed, the terminal orientation variable is allowed to vary over a much larger domain. Although this does not necessarily prevent the solver from finding a feasible incumbent, it weakens the relaxation and usually slows down the improvement of the best bound. As a result, the relative MIP gap may remain large for a longer period. In contrast, the exact equality constraint $\varphi(N) = 0$ provides stronger terminal information and can help tighten the formulation, which is beneficial for improving the best bound and reducing the gap once a feasible solution has been found. However, because the motion planning problem still contains multiple coupled mixed-integer constraints, an overly strict terminal condition may shrink the feasible region excessively and increase the difficulty of obtaining the first feasible incumbent.

The bounded constraint $\varphi(N) \in [-\pi/8,\pi/8]$ provides a more favorable compromise between these two effects. On the one hand, the tight upper and lower bounds offer informative domain restrictions for presolve and branch-and-bound, thereby improving bound quality. On the other hand, unlike the exact equality constraint, the inequality form preserves a limited but nonzero feasible region around the desired terminal heading, which makes it easier for the solver to identify an early feasible solution. Therefore, this setting achieves a better balance between parking accuracy and computational efficiency.

6. Real-world experiments

A prototype 4WID-4WS vehicle serves as the experimental platform to further validate the proposed algorithm. As shown in Fig. 26, the platform is equipped with four hub motors and two steer-by-wire actuators, enabling multiple maneuvering modes such as ASM, DMM, and ZRM. However, due to mechanical limitations of the prototype, closed-loop testing in ZRM cannot be conducted safely. To avoid potential damage to the prototype vehicle, only ASM and DMM are tested in real vehicle experiments. As an alternative, we assume the vehicle can perform ZRM at appropriate locations, which does not affect the validity of our conclusions. The vehicle's position, orientation, and velocity are recorded in real time using a navigation device. The software runs on an onboard industrial computer with an i9 ®Core™ processor and uses the Robot Operating System (ROS) as the software framework. Key parameters of the platform are listed in Table 7.

This study employs two realistic constrained scenarios to validate the proposed method. As shown in Fig. 27, the first is a dead-end parallel parking scenario, where the parking space is located at the road end with limited aisle width and restricted maneuvering space. The second is an obstructed exit scenario, where the vehicle is partially blocked by an illegally parked vehicle while exiting

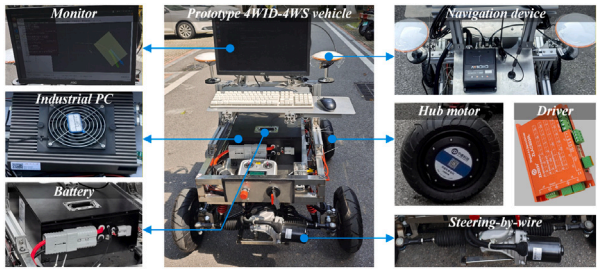
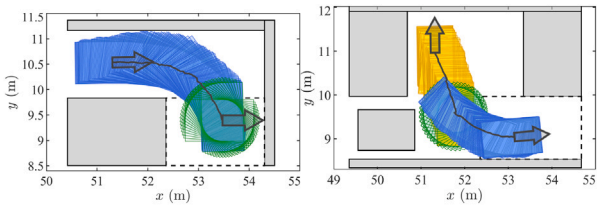


Fig. 26. 4WID-4WS experimental platform.



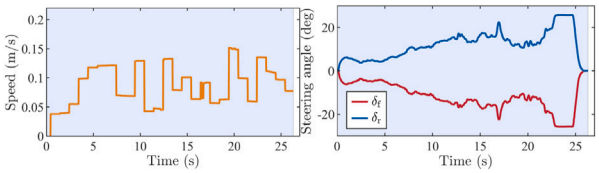
(a) Scenario 1 (b) Scenario 2

Fig. 27. Typical constrained scenarios selected for experiments. Blue arrows indicate the starting poses, and red boxes represent the target areas.



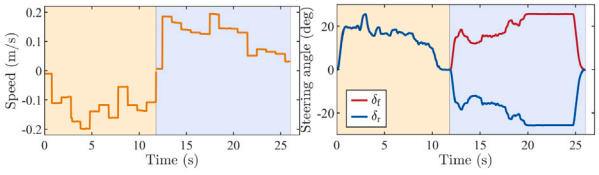
(a) Scenario 1 (b) Scenario 2

Fig. 28. Pose evolution of closed-loop trajectories. The ZRM trajectory is shown with interpolation for visualization.



(a) Speed of CoG (b) Steering angle

Fig. 29. Speed and wheel angle profiles under Scenario 1.



(a) Speed of CoG (b) Steering angle

Fig. 30. Speed and wheel angle profiles under Scenario 2. The blue and yellow regions represent the ASM and DMM, respectively.

and must maneuver out toward the right. Obstacles in both scenarios are represented as polygonal no-go zones and provided to the planner as perceptual inputs. Using a simple pure pursuit controller enables accurate closed-loop tracking of the planned trajectories under both ASM and DMM.

Fig. 28 shows the closed-loop trajectories for both scenarios. In Scenario 1, the vehicle first reaches the target area using ASM regardless of the final heading, then employs ZRM for orientation adjustment. This strategy avoids repeated forward-backward switching near the parking space. In Scenario 2, the vehicle initially exits using DMM to prevent sweeping against the left-side obstacle, then utilizes ZRM to change direction and complete the maneuver. Figs. 29 and 30 display the speed and steering angle profiles under ASM and DMM in the two scenarios. The experimental setup requires the ICR to lie on the perpendicular bisector of the longitudinal axis during ASM. Consequently, the front and rear steering angles are equal in magnitude but opposite in direction under ASM, while

they remain identical under DMM. The results demonstrate that the proposed method generates easily trackable trajectories, and the closed-loop motion successfully leverages multiple maneuvering modes to complete the tasks without collisions.

7. Conclusion

This work has presented a generic spatio-temporal decoupling framework for motion planning of MLAVs that integrates a modified A*-based coarse search, a path optimization with decision of maneuvering modes, and a segmented temporal layer with transient switching optimization. The modified A* produces an initial, multi-mode feasible path and provides guidance for the MIP. The MIP then refines path geometry while resolving discrete mode decisions to ensure kinematic feasibility. Finally, two intermediate switching models (ICR-based and simplified single-track) regularize transitions so the temporal optimizer can generate smooth switching profiles. Together these components form a coherent, tractable pipeline that recovers mobility lost to planning-oriented simplifications while keeping computation practical.

The proposed spatio-temporal decoupling framework offers a practical and theoretically consistent pipeline from optimal decision to smooth switching of maneuvering modes. It provides a new structured paradigm for MLAV motion planning, with demonstrated improvements in trajectory quality, flexibility, and efficiency in constrained environments. Future work will focus on enhancing the computational efficiency of the MIP stage and extending the framework to handle dynamic obstacles.

CRediT authorship contribution statement

Yuan Chang: Writing – review & editing, Writing – original draft, Software, Methodology. **Zeyu Yang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Manjiang Hu:** Supervision, Project administration, Funding acquisition. **Xianfeng Wang:** Resources, Funding acquisition. **Yougang Bian:** Software, Methodology, Conceptualization. **Chengqi Long:** Software, Methodology, Formal analysis. **Jianghua Feng:** Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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